Course Code: ICT-2101

Course Title: Operational Amplifier and Linear Integrated Circuits

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TEXTBOOKS:

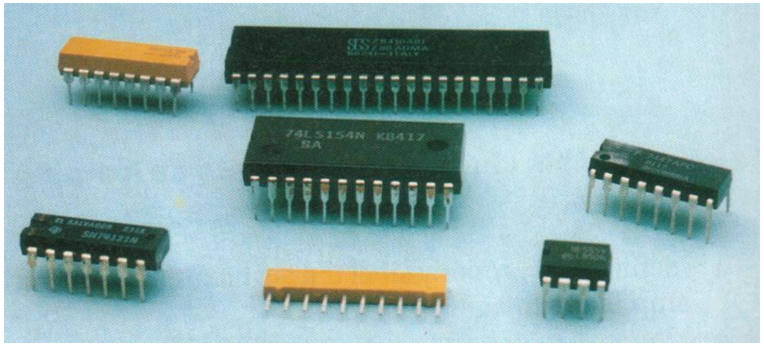
Ramakant A, Gayakward
"Op-Amps and Linear Integrated Circuits"
Prentice Hall of India, New Delhi, 4th Edition.

REFERENCES:

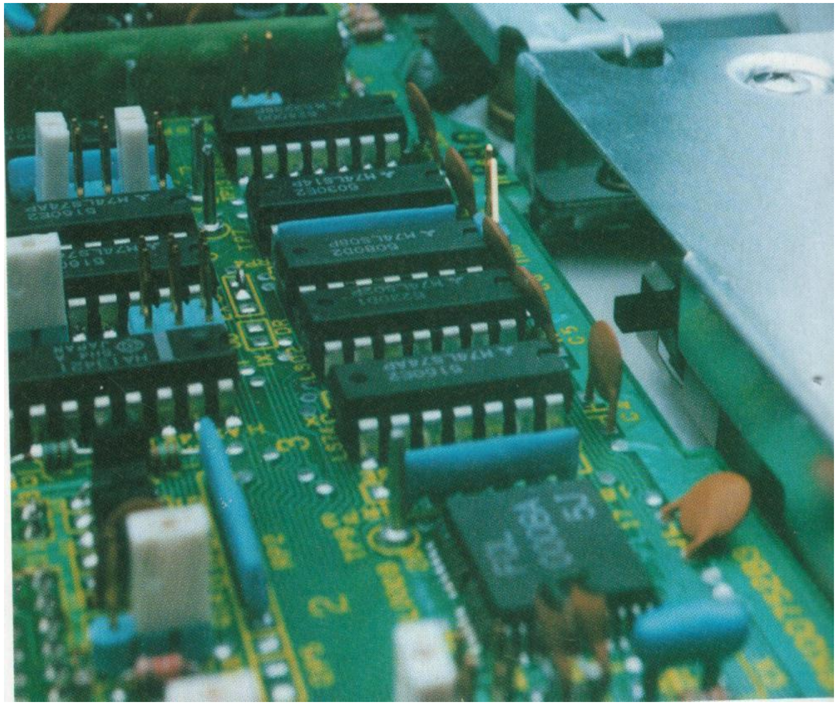
1. Behzad Razavi
"Design of Analog CMOS Integrated Circuits"
McGraw Hill, 2001.

2. D. Roy Choudhry, Shail Jain
"Linear Integrated Circuits"
New Age International Pvt. Ltd., 2000.

INTRODUCTION



Typical IC packages



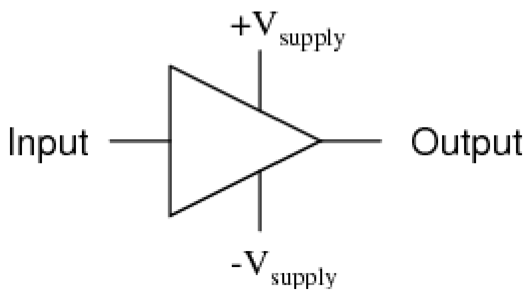
IC packages placed on circuit board

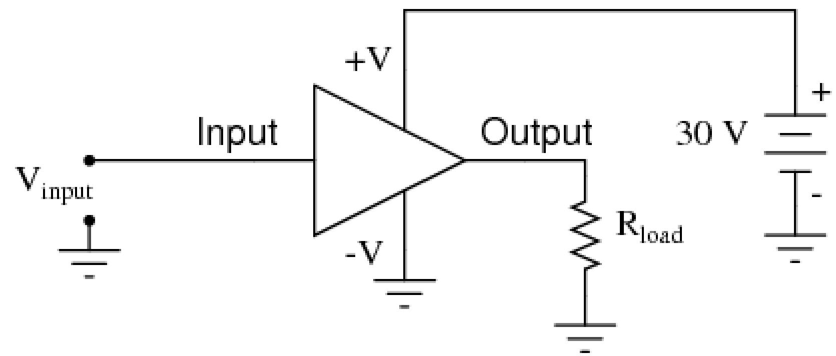
Definition

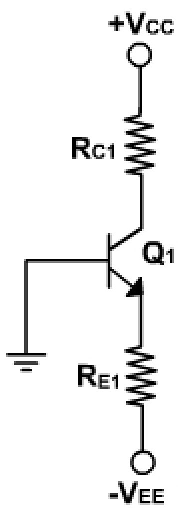
- The **operational amplifier** or op-amp is a circuit of components integrated into one chip.
- It is capable of sensing and amplifying dc and ac input signals
- A typical op-amp is powered by two dc voltages and has one inverting(-) input, one non-inverting input (+) and one output.
- The word "operational" is used because the amplifier can be used to perform a variety of mathematical operations such as addition, subtraction, integration, differentiation etc.

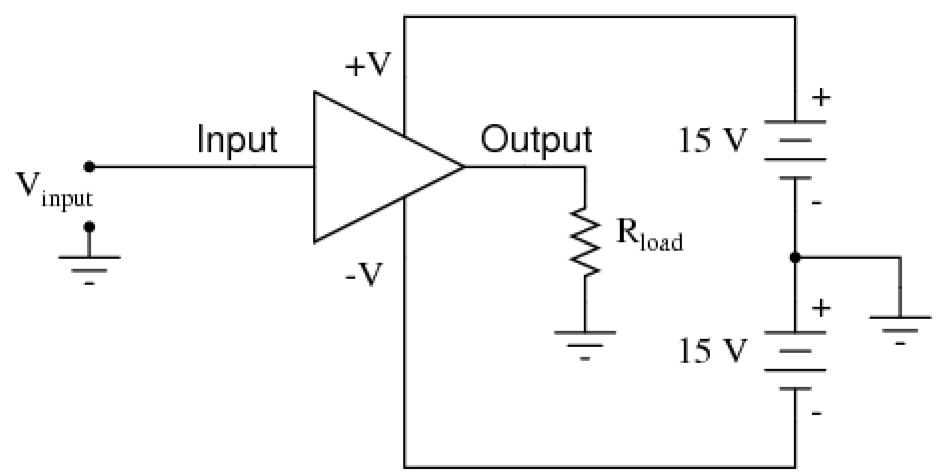
General Amplifier

General amplifier circuit symbol

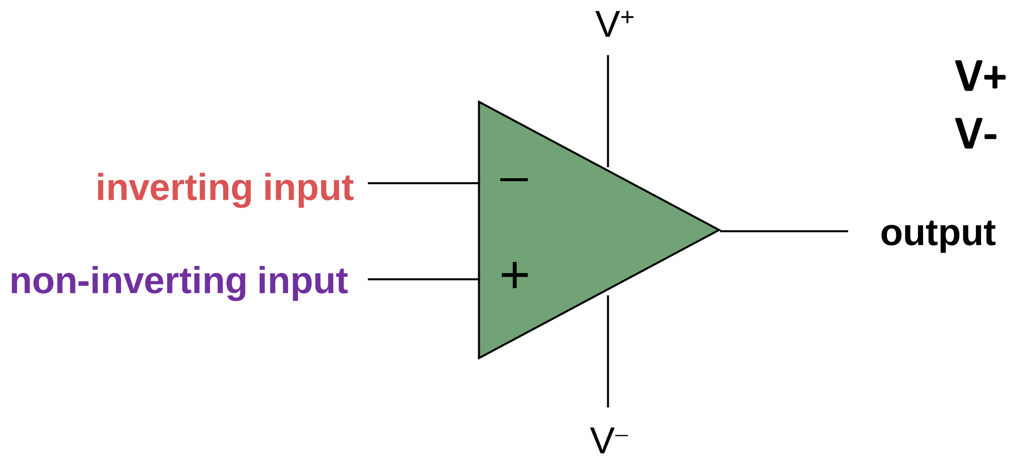








SYMBOL OF OPERATIONAL AMPLIFIER:



$V+ \leftrightarrow +V_{CC}$

$V- \leftrightarrow -V_{EE}$ or $-V_{cc}$

The Gain of OP-AMP is denoted by "A".

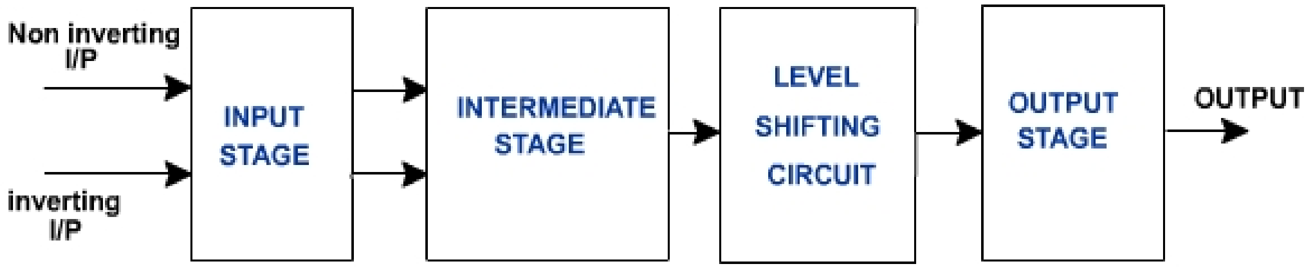
A = (Output/Difference Between Two Input Signals)

$= Vo /(V_1 – V_2)$

Where,

V_1 = Voltage Applied at Non-Inverting input
V_2 = Voltage Applied at Inverting input
V_o = Output Voltage

BLOCK DIAGRAM OF OP-AMP:



Input stage: It consists of a dual input, balanced output differential amplifier. Its function is to amplify the difference between the two input signals. It provides high differential gain, high input impedance and low output impedance.

Intermediate stage: The overall gain requirement of an Op-Amp is very high. Since the input stage alone cannot provide such a high gain. Intermediate stage is used to provide the required additional voltage gain.

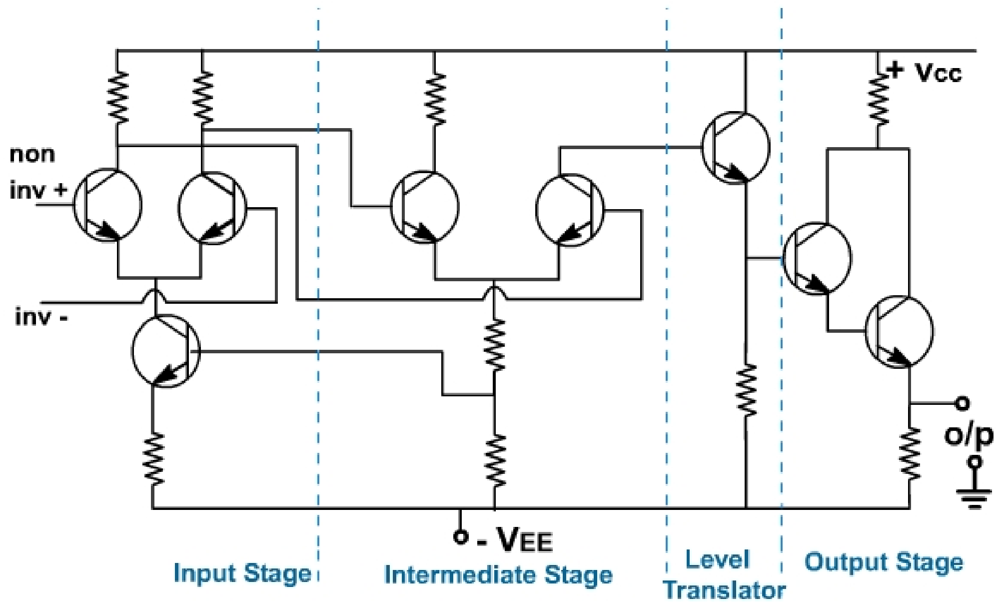
It consists of another differential amplifier with dual input, and unbalanced (single ended) output.

Block Diagram of OP-AMP:

Buffer and Level shifting stage: As the Op-Amp amplifies D.C signals also, the small D.C. quiescent voltage level of previous stages may get amplified and get applied as the input to the next stage causing distortion the final output.

Hence the level shifting stage is used to bring down the D.C. level to ground potential, when no signal is applied at the input terminals. Buffer is usually an emitter follower used for impedance matching.

Output stage: It consists of a push-pull complementary amplifier which provides large A.C. output voltage swing and high current sourcing and sinking along with low output impedance.



Definition: The "Integrated Circuit " or IC is a miniature, low cost electronic circuit consisting of active and passive components that are irreparably joined together on a single crystal chip of silicon.

Active components: FETs, and BJTs

Passive components: Capacitors, Inductors, and Resistors.

In 1958 Jack Kilby of Texas Instruments invented first IC

Applications of an Integrated Circuit

- Only a half century after their development was initiated, integrated circuits have become ubiquitous.

- Computers, cellular phones, and other digital appliances are now inextricable parts of the structure of modern societies.

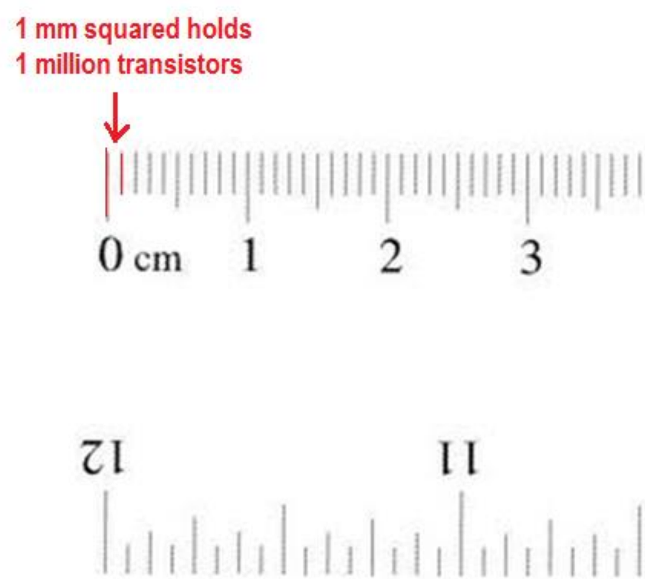
- Modern computing, communications, manufacturing and transport systems, including the Internet, all depend on the existence of integrated circuits.

ADVANTAGES:

- Small size
- Low cost
- Less weight
- Low supply voltages
- Low power consumption
- Highly reliable
- Matched devices
- Fast speed

- Performance is also enhanced by the stringent test requirements of ICs, the IC is 30 to 50 times more reliable than conventional circuitry.

- As of 2006, chip areas range from a few square mm to around 350 mm, with up to 1 million transistors per mm squared.



THIS RULER IS A
LITTLE BIT LARGER
THAN TRUE SCALE

- There are some disadvantages to ICs:

1. CAN'T HANDLE HIGH CURRENTS
2. CAN'T HANDLE HIGH VOLTAGES
3. LIMITED TO ONLY 4 COMPONENTS
4. CAN'T BE REPAIRED

- High current generates heat that can easily damage tiny components.

- High voltages can break down the insulation between components and the IC because the components are very close together.

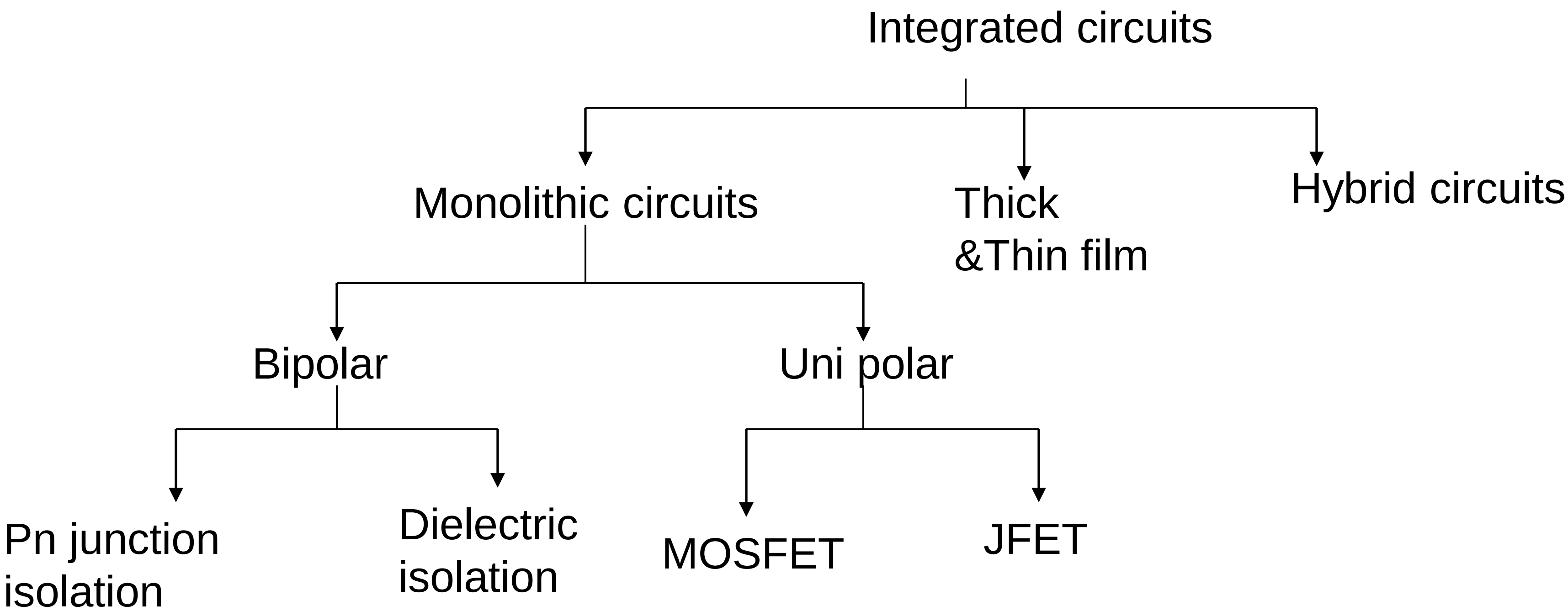
- There are only four components that can be constructed within the IC; transistors, diodes, capacitors and resistors.
- Transistors and diodes are the easiest to construct in an IC, however capacitors and resistors take up more room especially as their values increase.

- Integrated circuits can't be repaired because their internal components can't be separated.

Classification of IC's

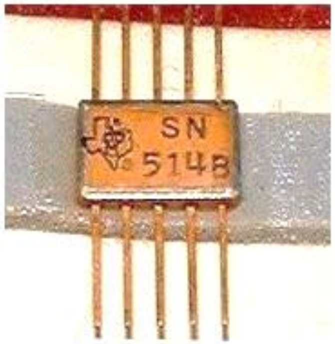
- On the basis of **fabrication techniques** used.
- On the basis of **the chip size**
- On the basis of **applications**

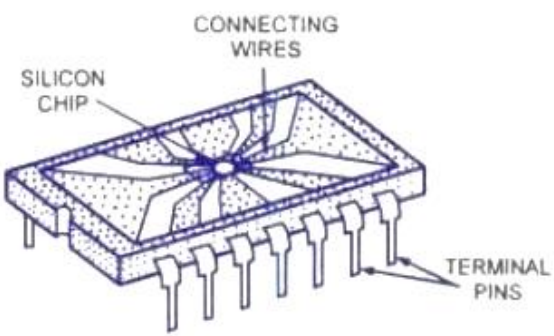
ON BASIS OF FABRICATION



<u>Classification of ICs</u>

MONOLITHIC IC'S

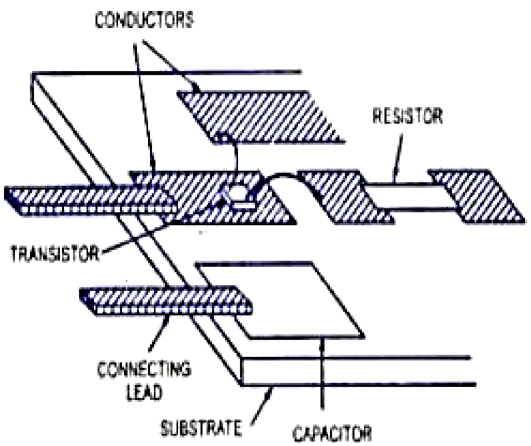




Monolithic IC in Plastic Package

Monolithic circuit is built into a single stone or single crystal i.e. in monolithic ICs, all circuit components, and their interconnections are formed into or on the top of a single chip of silicon. Monolithic ICs are by far the most common type of ICs used in practice, because of **mass production** , **lower cost** and **higher reliability**.

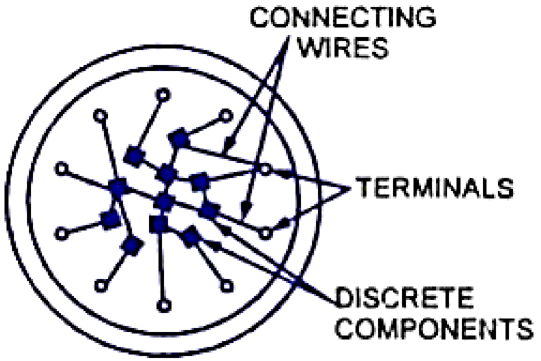
THIN AND THICK FILM IC'S



Enlarged Portion of Thick-Film IC

These devices are larger than monolithic ICs but smaller than discrete circuits. These ICs can be used when power requirement is comparatively higher. With a thin-or thick-film IC, the passive components like resistors and capacitors are integrated, but the transistors and diodes are connected as discrete components to form a complete circuit. The essential difference between the thin- and thick-film ICs is not their relative thickness but the method of deposition of film. Both have similar appearance, properties and general characteristics.

HYBRID IC'S



Hybird or Multichip IC

The circuit is fabricated by interconnecting a number of individual chips.

Hybrids ICs are widely used for high power **audio amplifier** applications .

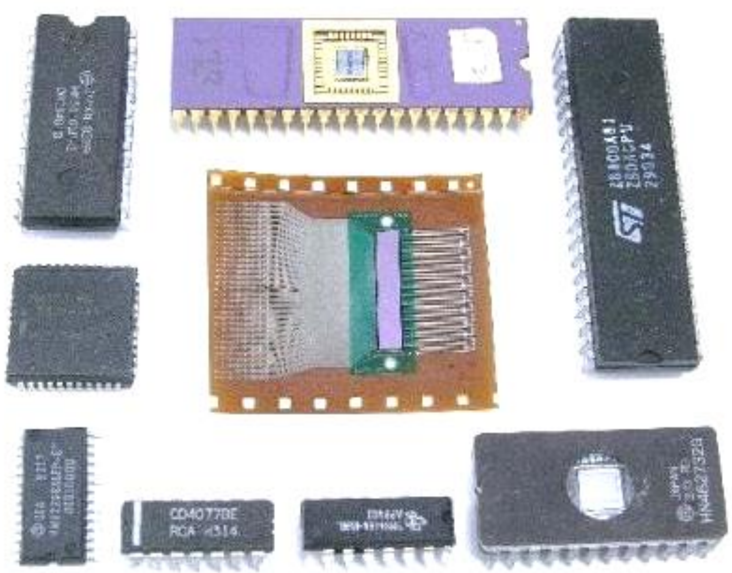
Have better performance than monolithic ICs

Process is too expensive for mass production

On basis of Chip size and Complexity

- Invention of Transistor (Ge) - 1947
- Development of Silicon - 1955-1959
- Silicon Planar Technology - 1959
- First ICs, SSI (3- 30gates/chip) - 1960
- MSI (30-300 gates/chip) - 1965-1970
- LSI (300-3000 gates/chip) -1970-1975
- VLSI (More than 3k gates/chip) - 1975
- ULSI (more than one million active devices are integrated on single chip)

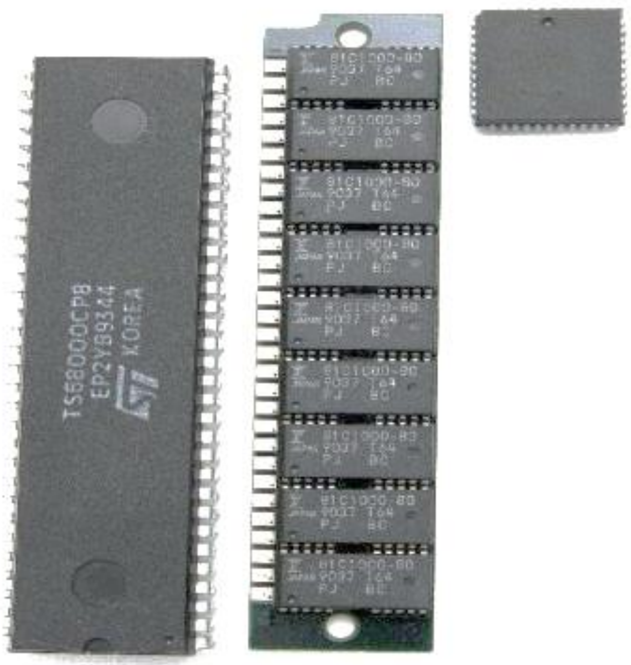
SSI AND MSI



Small scale integration (SSI) has 3 to 30 gates/chip orUp to 100 electronic components per chip

Medium scale integration (MSI) has 30 to 300 gates/chip or 100 to 3,000 electronic components per chip

LSI AND VLSI



Large scale integration (LSI)-300 to 3,000 gates/chip or 3,000 to 100,000 electronic components per chip

Very large scale integration (VLSI)-more than 3,000 gates/chip or 100,000 to 1,000,000 electronic components per chip

ULSI

Ultra Large-Scale Integration (ULSI)- More than 1 million electronic components per chip The Intel 486 and Pentium microprocessors, for example, use ULSI technology. The line between VLSI and ULSI is vague.

ON BASIS OF APPLICATIONS

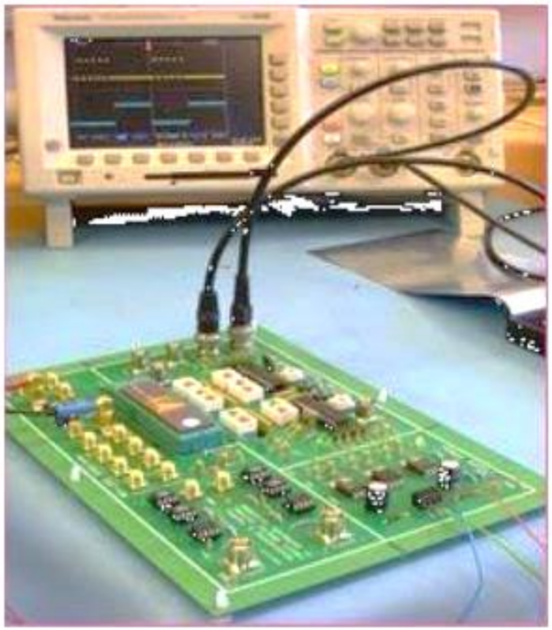
- *LINEAR INTEGRATED CIRCUITS*
- *DIGITAL INTEGRATED CIRCUITS*

LINEAR INTEGRATED CIRCUITS

When the input and output relationship of a circuit is linear, ***linear ICs*** are used. Input and output can take place on a continuous range of values.

Example operational amplifiers, power amplifiers, microwave amplifiers multipliers etc.

DIGITAL INTEGRATED CIRCUITS



When the circuit is either in on-state or off-state and not in between the two, the circuit is called the digital circuit. ICs used in such circuits are called the ***digital ICs***. They find wide applications in computers and logic circuits.
Example logic gates, flip flops, counters, microprocessors, memory chips etc.

SSI	MSI	LSI	VLSI	ULSI
< 100 active devices	100-1000 active devices	1000-100000 active devices	>100000 active devices	Over 1 million active devices
Integrated resistors, diodes & BJT's	BJT's and Enhanced MOSFETS	MOSFETS	8bit, 16bit Microprocessors	Pentium Microprocessors

IC Package types:

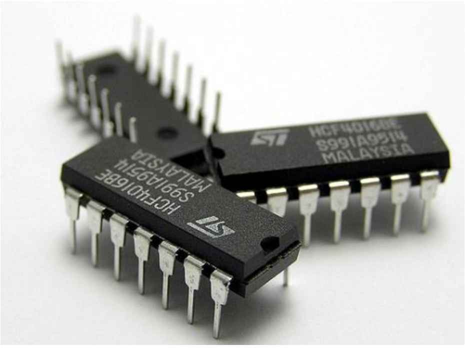
- Metal can Package
- Dual-in-line
- Flat Pack

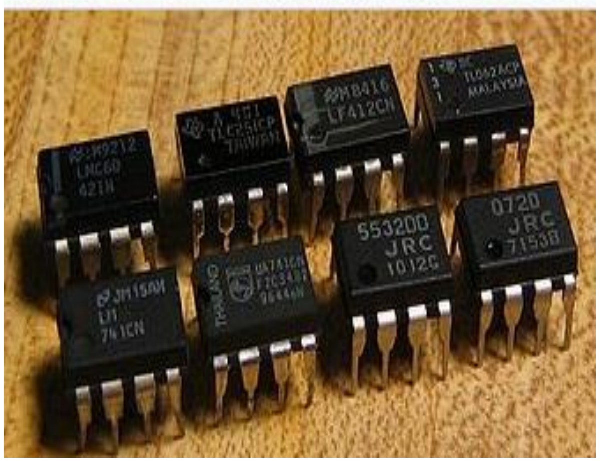
Metal can Packages

- The metal sealing plane is at the bottom over which the chip is bounded
- It is also called **transistor pack**

Dual-in-line Package

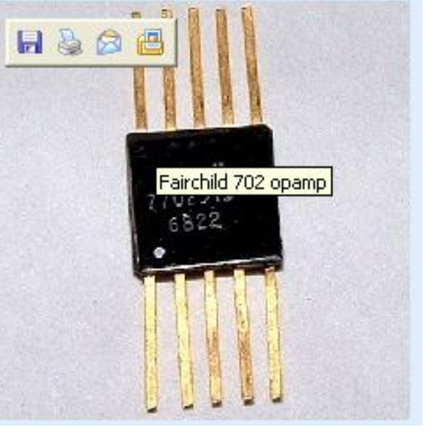
- The chip is mounted inside a plastic or ceramic case
- The 8 pin Dip is called **Mini-DIP** and also available with 12, 14, 16, 20pins





FLAT PACK

- The chip is enclosed in a rectangular ceramic case

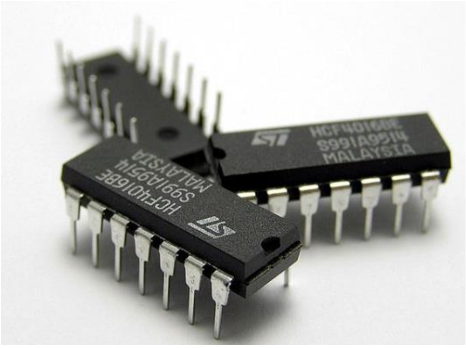


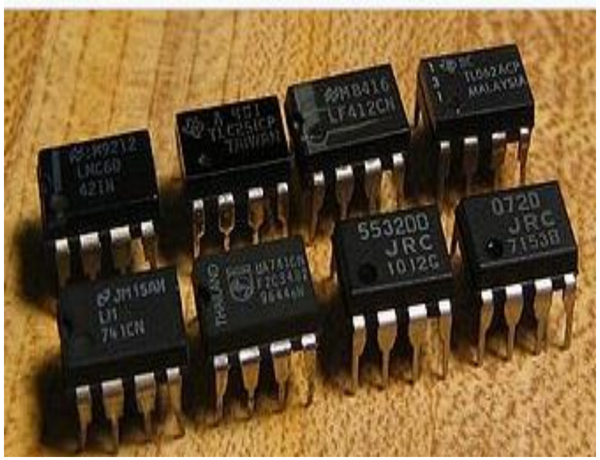
Selection of IC Package

Type	Criteria
Metal can package	1. Heat dissipation is important 2. For high power applications like power amplifiers, voltage regulators etc.
DIP	1. For experimental or bread boarding purposes as easy to mount 2. If bending or soldering of the leads is not required 3. Suitable for printed circuit boards as lead spacing is more
Flat pack	1. More reliability is required 2. Light in weight 3. Suited for airborne applications

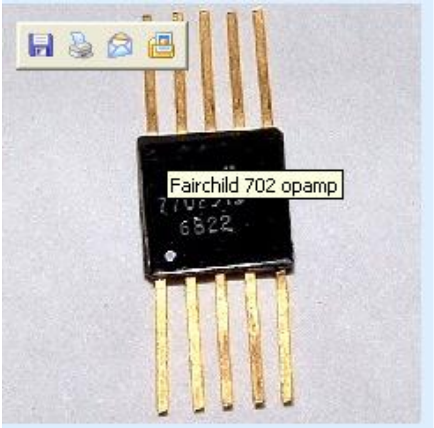
Packages

The metal can (TO) Package





The Dual-in-Line (DIP) Package



The Flat Package

Factors affecting selection of IC package

- Relative cost
- Reliability
- Weight of the package
- Ease of fabrication
- Power to be dissipated
- Need of external heat sink

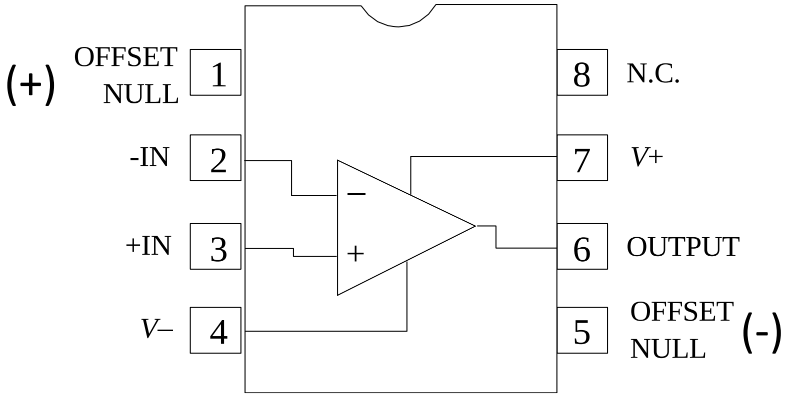
Temperature Ranges

1. Military temperature range : -55^{o} C to $+125^{o}$ C (-55^{o} C to $+85^{o}$ C)
2. Industrial temperature range : -20^{o} C to $+85^{o}$ C (-40^{o} C to $+85^{o}$ C)
3. Commercial temperature range: 0^{o} C to $+70^{o}$ C (0^{o} C to $+75^{o}$ C)

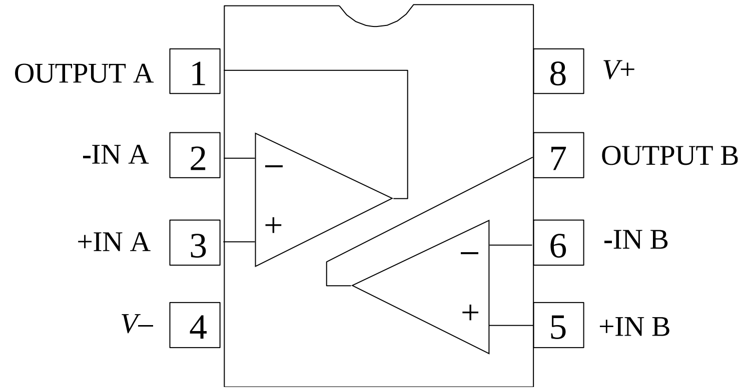
Manufacturer's Designation for Linear ICs

Fairchild	- µA, µAF
National Semiconductor	- LM,LH,LF,TBA
Motorola	- MC,MFC
RCA	- CA,CD
Texas Instruments	- SN
Signetics	- N/S,NE/SE
Burr- Brown	- BB

OP-AMP ICs:

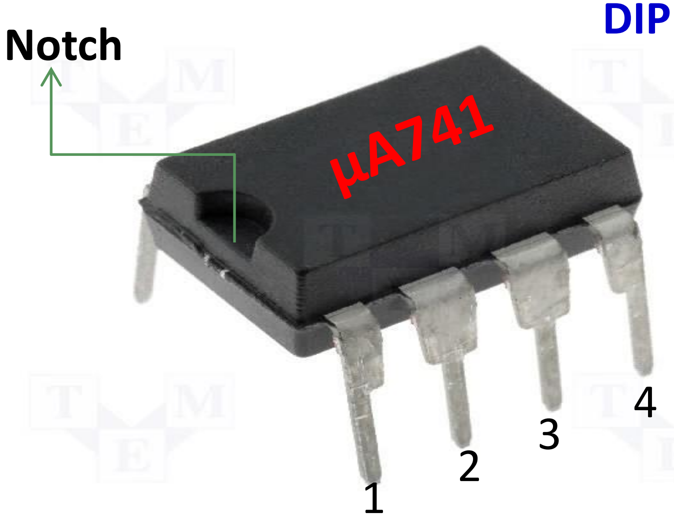


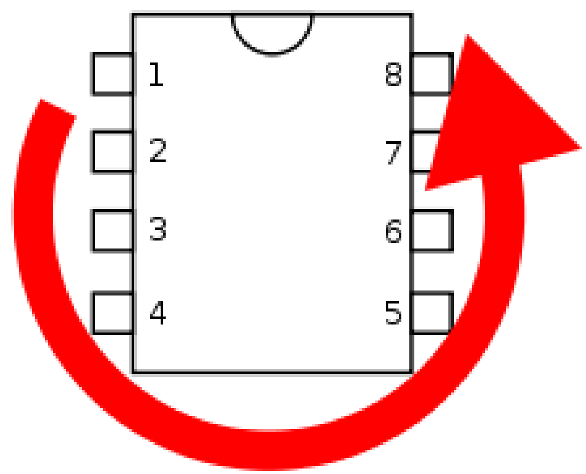
DIP-741



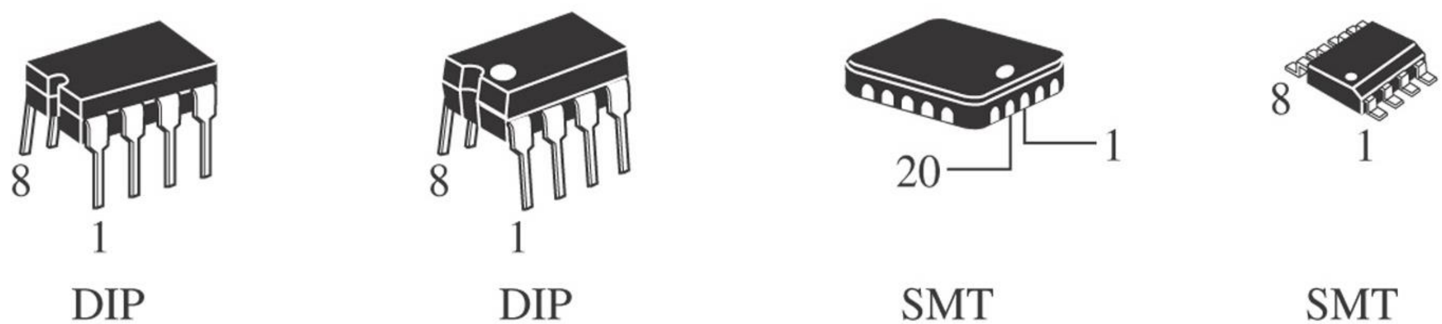
Dual op-amp 1458 device

DIP: Dual inline package



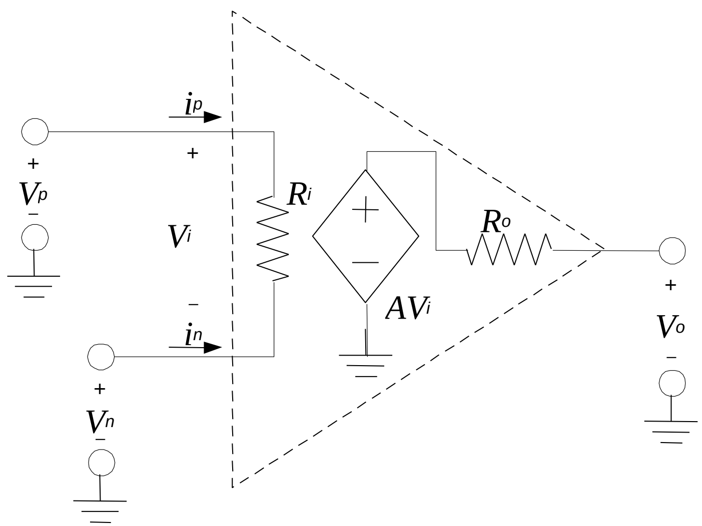


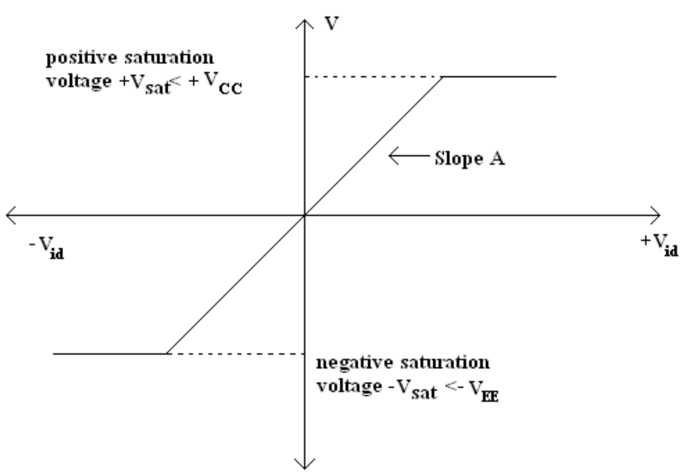
- Typical op-amp packages

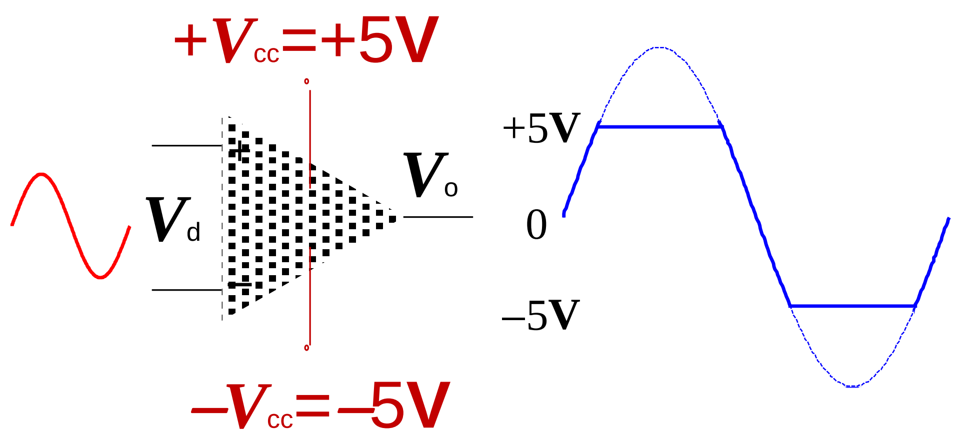


(c) Typical packages. Pin 1 is indicated by a notch or dot on dual in-line (DIP) and surface-mount technology (SMT) packages, as shown.

EQUIVALENT CIRCUIT OF OP-AMP & TRANSFER CHARACTERISTICS:







DIFFERENTIAL AND COMMONMODE OPERATION :

Differential Inputs: When separate inputs are applied to the op-amp, the resulting difference signal is the difference between the two inputs.

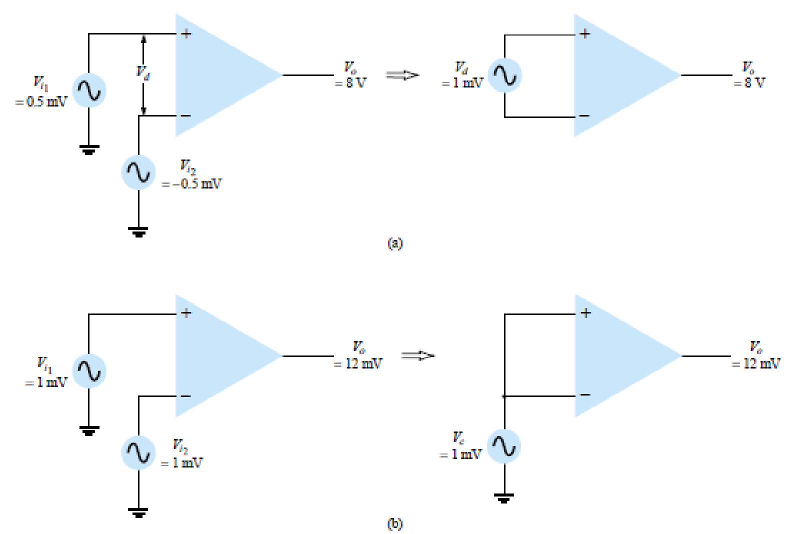
$$\boxed{V_d = V_{i_1} - V_{i_2}}$$

Common Inputs: When both input signals are the same, a common signal element due to the two inputs can be defined as the average of the sum of the two signals.

$$\boxed{V_c = \tfrac{1}{2}(V_{i_1} + V_{i_2})}$$

Output Voltage: Since any signals applied to an op-amp in general have both in-phase and out-phase components, the resulting output can be expressed as-

$$\boxed{V_o = A_d V_d + A_c V_c}$$



Where,

V_d: difference voltage

V_c: common voltage

A_d: differential gain of the amplifier

A_c: common-mode gain of the amplifier

CHARACTERISTICS AND PERFORMANCE PARAMETERS OF OP-AMP :

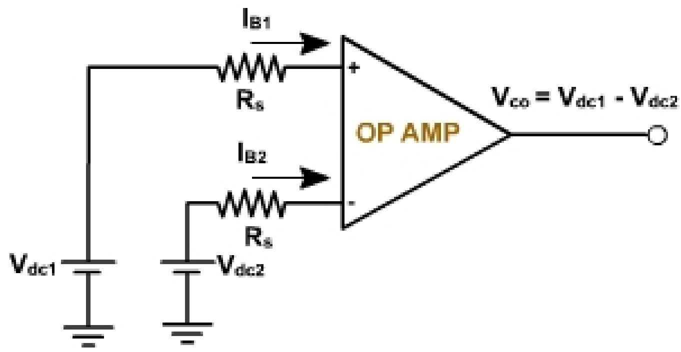
- **Input offset Voltage**
- **Input offset current**
- **Input bias current**
- **Differential input resistance**
- **Input capacitance**
- **Open loop voltage gain**
- **CMRR**
- **Output voltage swing**

CHARACTERISTICS AND PERFORMANCE PARAMETERS OF OP-AMP :

- **Output resistance**
- **Offset adjustment range**
- **Input Voltage range**
- **Power supply rejection ratio**
- **Power consumption**
- **Slew rate**
- **Gain – Bandwidth product**
- **Equivalent input noise voltage and current**
- **Average temperature coefficient of offset parameters**
- **Output offset voltage**
- **Supply current**

CHARACTERISTICS AND PERFORMANCE PARAMETERS OF OP-AMP :

Input offset Voltage : Input offset voltage is the voltage that must be applied between the two input terminals of an op-amp to null the output.



$$V_{io} = V_{dc1} - V_{dc2}$$

For op-amp 741C the input offset voltage is 6mV

Input offset Current: The algebraic difference between the current in the inverting and non inverting terminal is known as the input offset current $I_{io} = (I_{B1} - I_{B2})$.

For op-amp 741C the input offset current is 200nA

CHARACTERISTICS AND PERFORMANCE PARAMETERS OF OP-AMP :

Input Bias Current: I_B is the average current flows in the inverting and non-inverting terminal of an op-amp.

$$I_B = (I_{B1} + I_{B2})/2$$

For 741C the maximum value of I_b is 500nA

Differential Input Resistance: It is the equivalent resistance measured at either the inverting or non-inverting input terminal with the other input terminal grounded. It is denoted as R_i. For 741C it is of the order of 2MΩ

Input Capacitance: It is the equivalent capacitance measured at either the inverting or non- inverting input terminal with the other input terminal grounded. It is denoted as C_i. For 741C it is of the 1-4 pF.

CHARACTERISTICS AND PERFORMANCE PARAMETERS OF OP-AMP :

Open Loop Voltage gain: It is the ratio of output voltage to the differential input voltage, when op-amp is in open loop configuration, without any feedback. It is also called as large signal voltage gain

A = Output voltage/Differential input

$= V_o/V_{id}$

For 741C it is typically 200,000

Common Mode Rejection Ratio(CMRR): When the same voltage is applied to both the input terminals the voltage is called a common mode voltage V_{cm} and the op-amp is said to be operating in the common mode configuration,CMRR is defined as the ratio of the differential voltage gain to common mode gain.

CMRR = A_d/A_{cm}

$$\boxed{V_o = A_d V_d\left(1 + \frac{1}{\text{CMRR}}\frac{V_c}{V_d}\right)}$$

CHARACTERISTICS AND PERFORMANCE PARAMETERS OF OP-AMP :

Output offset Voltage : Output offset voltage is the output voltage when both the input terminals are grounded.

Output Voltage Swing: The op-amp output voltage gets saturated at +Vcc and –VEE and it cannot produce output voltage more than +Vcc and –VEE. Practically voltages +Vsat and –Vsat are slightly less than +Vcc and –VEE .

For op-amp 741C the saturation voltages are $\pm$ 13V for supply voltages $\pm$ 15V

Output Resistance: It is the equivalent resistance measured between the output terminal of the op-amp and ground. It is denoted as R_o. For op-amp 741 it is 75Ω

Offset voltage adjustment range: The range for which input offset voltage can be adjusted using the potentiometer so as to reduce output to zero. For op-amp 741C it is $\pm$ 15mV.

CHARACTERISTICS AND PERFORMANCE PARAMETERS OF OP-AMP :

Input voltage range: It is the range of common mode voltages which can be applied for which op-amp functions properly and given offset specifications apply for the op-amp. For $\pm$ 15V supply voltages, the input voltage range is $\pm$ 13V

Supply voltage Rejection Ratio: The change in an op-amps input offset voltage V_{io} caused by variations in the supply voltage is called the SVRR. It is expressed in microvolts per volt or in decibels.

$$SR = \Delta V_{io} / \Delta V \quad (\mu V/V \text{ or dB})$$

The typical value of PSRR for op-amp 741C is 30µV/V

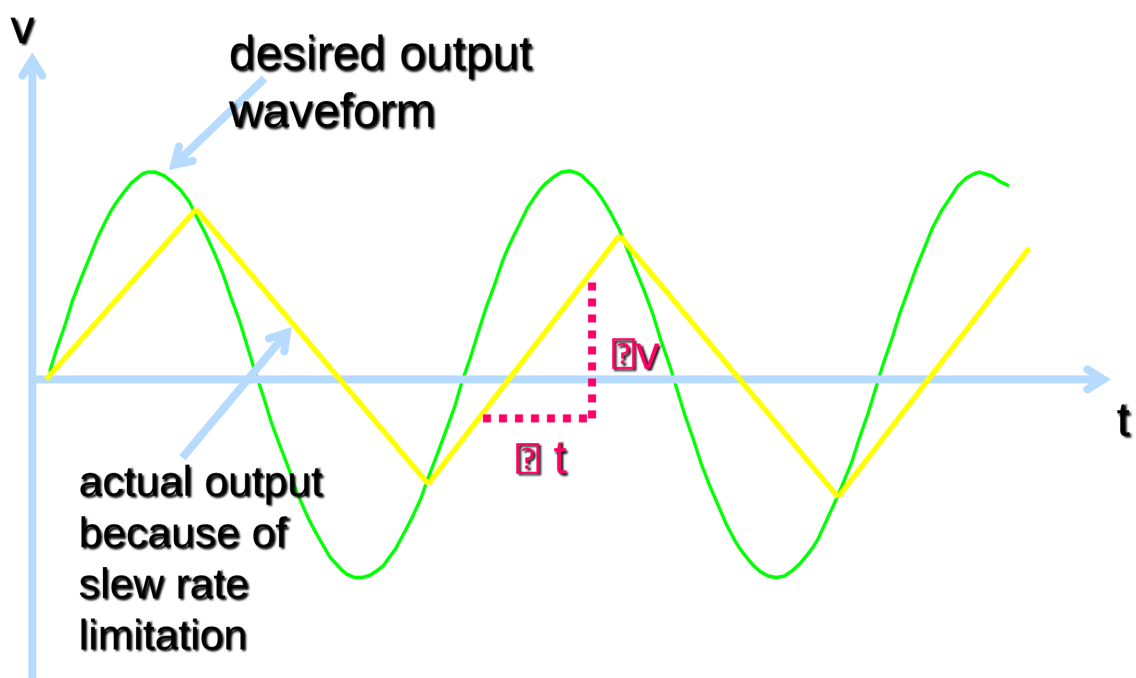
Power Consumption: It is the amount of quiescent power to be consumed by op-amp with zero input voltage, for its proper functioning. It is denoted as P_c

For 741C it is 85mW

CHARACTERISTICS AND PERFORMANCE PARAMETERS OF OP-AMP :

Slew Rate: Slew rate is defined as the maximum rate of change of output voltage per unit of time and is expressed as volt per micro second.

SR=(I dVo/dt I)max (V/µs)



For 741 IC the charging current is 15 µA and the internal capacitor is 30 pF. S= 0.5V/ µsec

Slew rate equation

$$V_s = V_m \sin\omega t$$

$$V_o = V_m \sin\omega t$$

$$\frac{dVo}{dt} = V_m\,\omega \cos\omega t$$

$$S = \text{slew rate} = \left.\frac{dVo}{dt}\right|_{max}$$

$$S = V_m\,\omega = 2\,\pi\, f\, V_m$$

$$\boxed{S = 2\,\pi\, f\, V_m \ V/sec}$$

This is also called **full power bandwidth** of the op-amp

For distortion free output, the maximum allowable input frequency f_m can be obtained as

$$\boxed{f_m = \frac{S}{2\Pi V_m}}$$

OP-AMP Parameters:

Gain Bandwidth Product : The gain bandwidth product(GB) is the bandwidth of the op-amp when the voltage gain is 1.

Ideal Vs Practical Parameters of OP-AMP :

	Ideal	**Practical (µA741)**
Open Loop gain A	$\propto$	2 X 10^5
Input Impedance Z_{in}	$\propto$	2 MΩ
Output Impedance Z_{out}	0	75 Ω
Input offset Voltage	0	2 mV
Input offset Current	0	200 nA
Bandwidth BW	$\propto$	1MHz
CMRR	$\propto$	90dB
Slew Rate	$\propto$	0.5 V/µSec.

Note: Please Remember all these values for µA741.

Q. Determine the output voltage of an op-amp for input voltages of V_{i1} 150 V, V_{i2} 140 V. The amplifier has a differential gain of Ad=4000 and the value of CMRR is: (a) 100. (b) 105.

Ans:

(a)

$$V_d = V_{i_1} - V_{i_2} = (150 - 140)\ \mu\text{V} = 10\ \mu\text{V}$$

$$V_c = \frac{1}{2}(V_{i_1} + V_{i_2}) = \frac{150\ \mu\text{V} + 140\ \mu\text{V}}{2} = 145\ \mu\text{V}$$

$$V_o = A_d V_d\left(1 + \frac{1}{\text{CMRR}}\frac{V_c}{V_d}\right)$$

$$= (4000)(10\ \mu\text{V})\left(1 + \frac{1}{100}\frac{145\ \mu\text{V}}{10\ \mu\text{V}}\right)$$

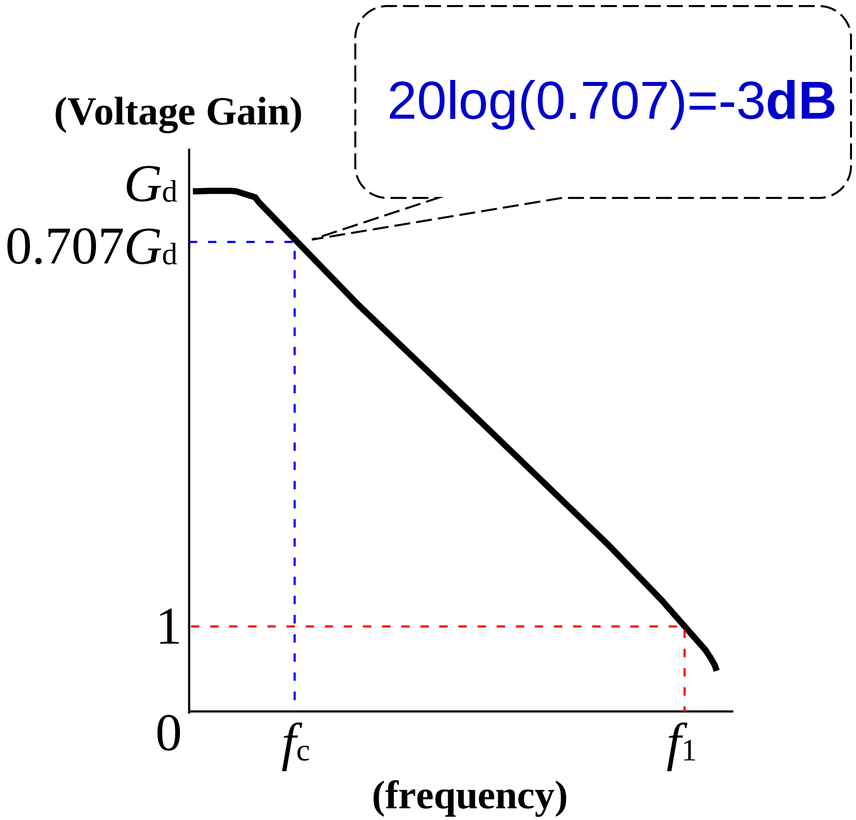
$$= 40\ \text{mV}(1.145) = \mathbf{45.8\ mV}$$

(b) $V_o = (4000)(10\ \mu\text{V})\left(1 + \frac{1}{10^5}\frac{145\ \mu\text{V}}{10\ \mu\text{V}}\right) = 40\ \text{mV}(1.000145) = \mathbf{40.006\ mV}$

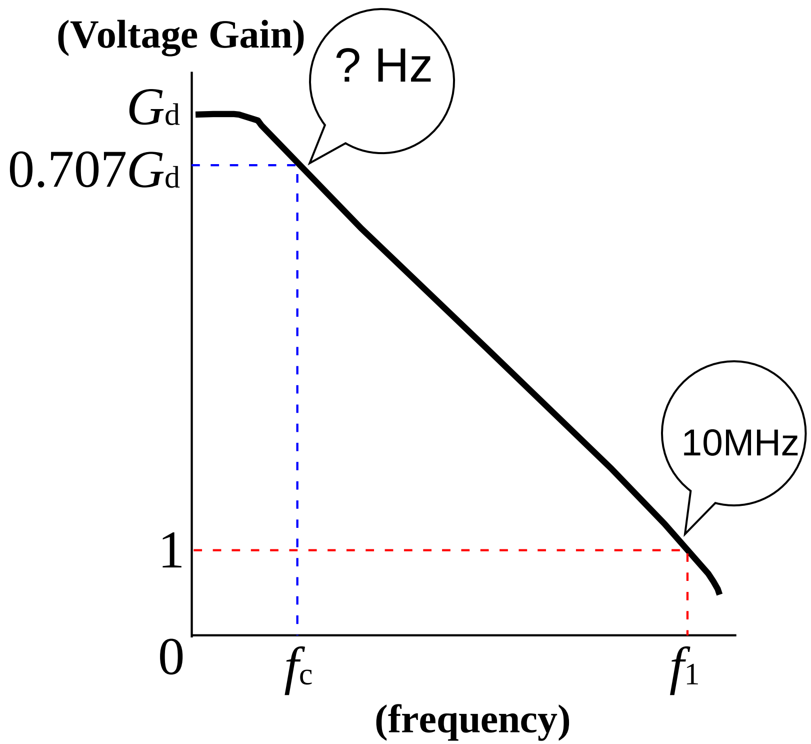
Frequency-Gain Relation:

- Ideally, signals are amplified from DC to the highest AC frequency
- Practically, bandwidth is limited
- 741 family op-amp have an limit bandwidth of few KHz.
- Unity Gain frequency f1: the gain at unity
- Cutoff frequency f_c: the gain drop by 3dB from dc gain G_d.

$$\boxed{\text{GB Product}: f_1 = G_d f_c}$$



Q. Determine the cutoff frequency of an op-amp having a unit gain frequency f_1 = 10 MHz and voltage differential gain G_d = 20V/mV.



Sol:

Since f_1 = 10 MHz

By using GB production equation

$f_1 = G_d\, f_c$

$f_c = f_1 / G_d$ = (10 MHz) / (20 V/mV)

= $(10 \times 10^6) / (20 \times 10^3)$

= 500 Hz

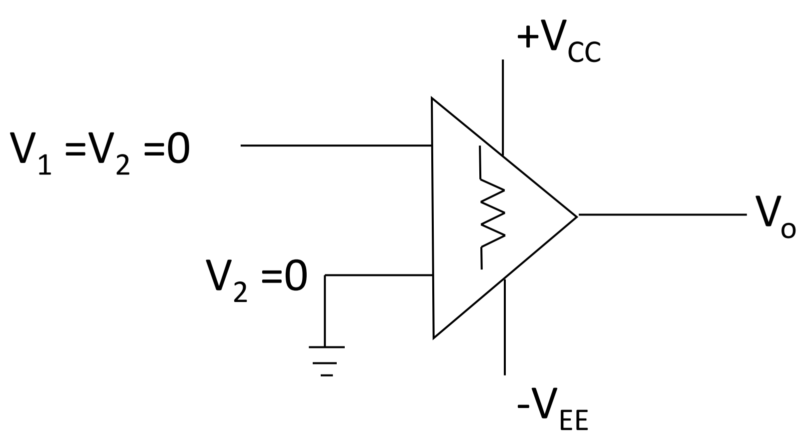
Concept of Virtual ground:

We know that, an ideal Op-Amp has perfect balance (ie output will be zero when input voltages are equal).

Hence when output voltage Vo = 0, we can say that both the input voltages are equal ie $V_1 = V_2$.

Since the input impedances of an ideal Op-Amp is infinite (Ri = inf). There is no current flow between the two terminals.

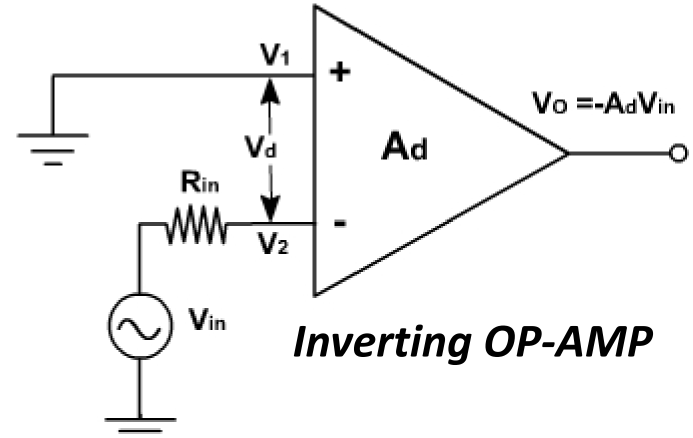
Hence when one terminal (say V2) is connected to ground (ie V2 = 0) as shown. Then because of virtual ground V1 will also be zero.



Open Loop OP-AMP Configuration :

There are three Open loop Configuration Namely:

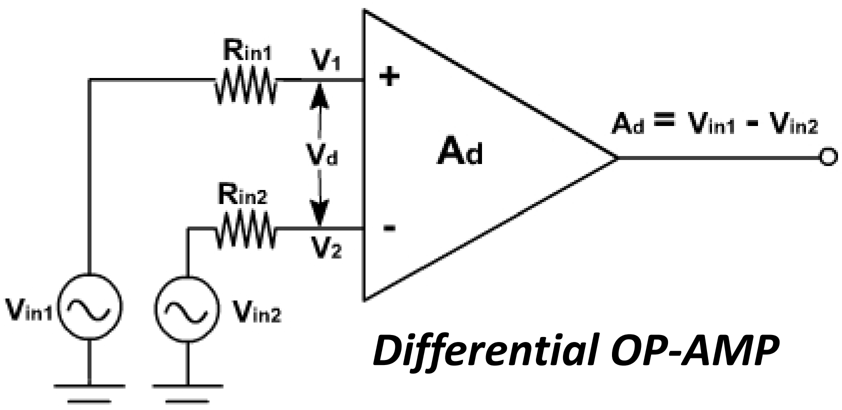
1. Differential OP-AMP
2. Inverting OP-AMP
3. Non-inverting OP-AMP



Inverting OP-AMP

$v_1 = 0$, $v_2 = v_{in}$.

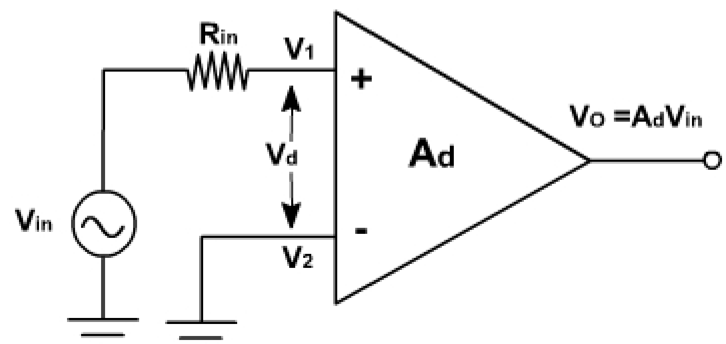
$v_o = -A_d\, v_{in}$



Differential OP-AMP

$v_1 = v_{in1}$ and $v_2 = v_{in2}$.

$v_o = A_d\,(v_{in1} – v_{in2}\,)$



Non-Inverting OP-AMP

$v_1 = +v_{in}\ ,\ v_2 = 0$

$v_o = +A_d\, v_{in}$

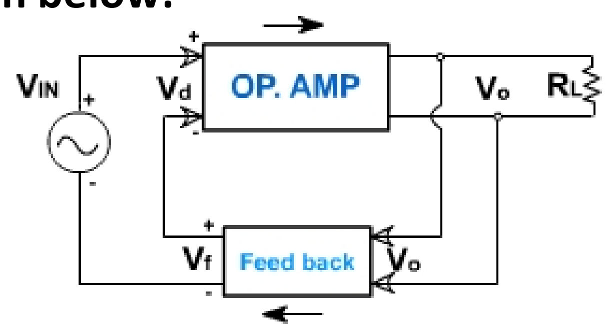
OP-AMP with Feedback :

Since OP-AMP gain A is very large, thus when operated in open loop, the OP-AMP goes into saturation.

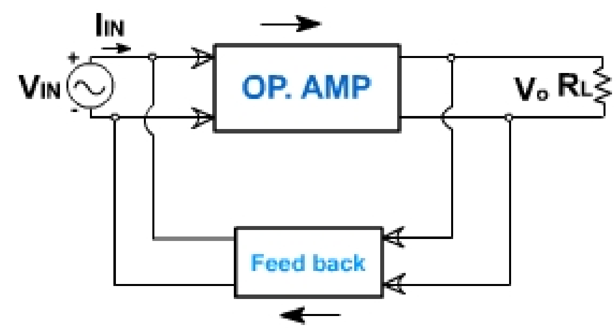
The gain of amplifier can be controlled by modifying the basic circuit i.e. adding feedback. Generally the negative feedback is added, which reduces the gain and also added other advantages.

A closed loop amplifier can be represented by two blocks one for an OP-AMP and other for a feedback circuits. There are four following ways to connect these blocks. The Basic configuration of feedback is given below:

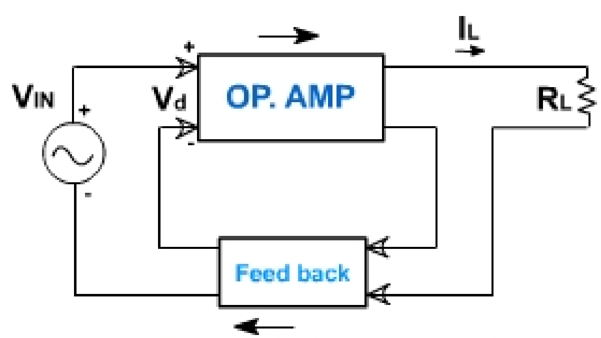
1. Voltage – series feedback
2. Voltage – shunt feedback
3. Current – series feedback
4. Current – shunt feedback



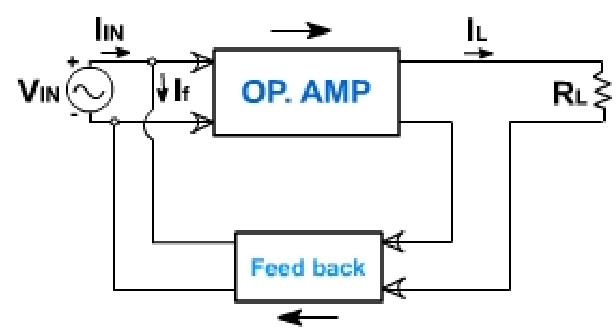
Voltage series feedback



Voltage shunt feedback



Current series feedback



Current shunt feedback

Op-Amp "Golden Rules"

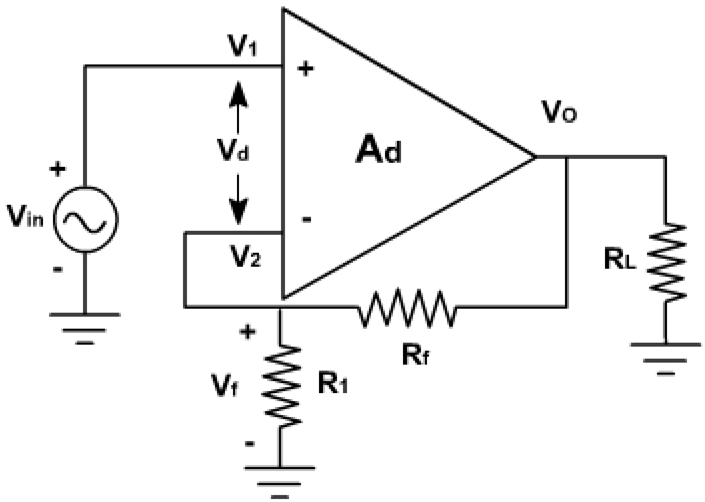
When an op-amp is configured in any negative-feedback arrangement, it will obey the following two rules:

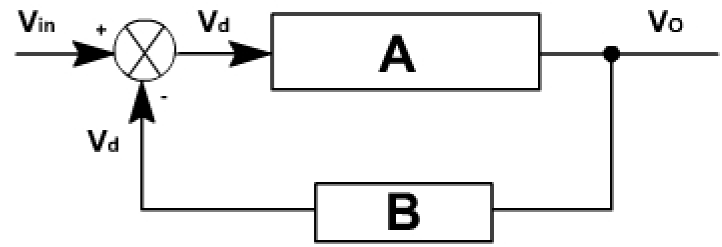
- ✓ **The inputs to the op-amp draw or source no current (true whether negative feedback or not)**

- ✓ **The op-amp output will do whatever it can (within its limitations) to make the voltage difference between the two inputs zero.**

Voltage series feedback:

It is also called non-inverting voltage feedback circuit. With this type of feedback, the input signal drives the non-inverting input of an amplifier; a fraction of the output voltage is then fed back to the inverting input. The op-amp is represented by its symbol including its large signal voltage gain Ad or A, and the feedback circuit is composed of two resistors R1 and Rf.





Open loop voltage gain $A_d = \dfrac{v_o}{v_d}$

Closed loop voltage gain $A_{CL} = \dfrac{v_o}{v_{in}}$

Feedback circuit gain $B = \dfrac{v_f}{v_o}$

The differential voltage input $v_d = v_{in} - v_f$

$$v_1 = v_{in}$$

$$v_2 = v_f = \frac{R_1}{R_1 + R_f} v_o$$

$$\therefore\ v_1 = v_2$$

$$\therefore\ v_{in} = \frac{R_1}{R_1 + R_f} v_o$$

$$v_o = \left(1 + \frac{R_1}{R_f}\right) v_{in}$$

$$A_{CL} = \frac{v_o}{v_{in}}$$

$$v_o = A\,(v_1 - v_2) = A\,(v_{in} - v_f)$$

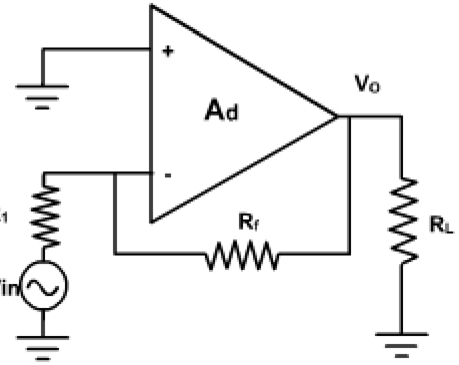
$$= A\,(v_{in} - Bv_o)$$

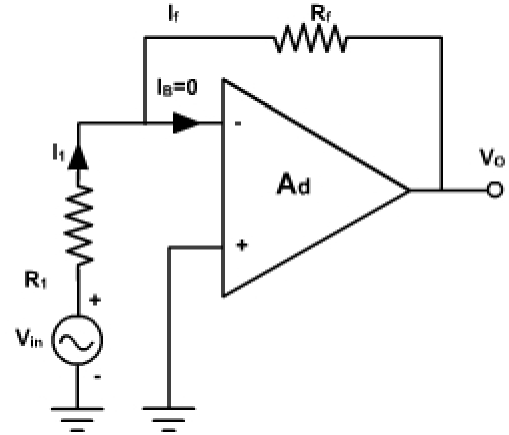
$$v_o\,(1 + AB) = A\,v_{in}$$

$$\frac{v_o}{v_{in}} = \frac{A}{1 + AB}$$

Voltage shunt Feedback:

It is also called inverting voltage feedback circuit.





$$i_{in} \simeq i_f$$

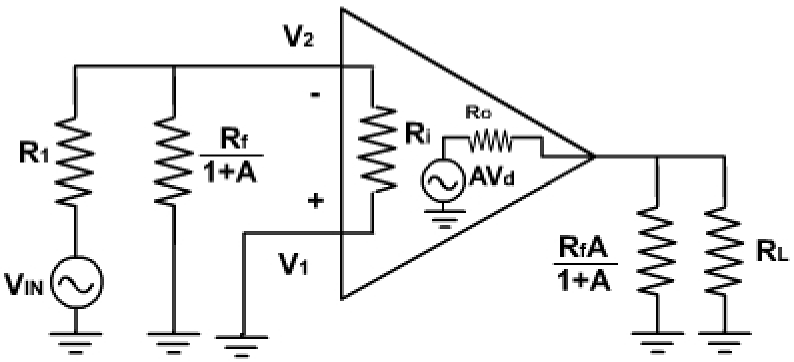
$$\frac{v_{in} - v_2}{R_1} = \frac{v_2 - v_o}{R_f}$$

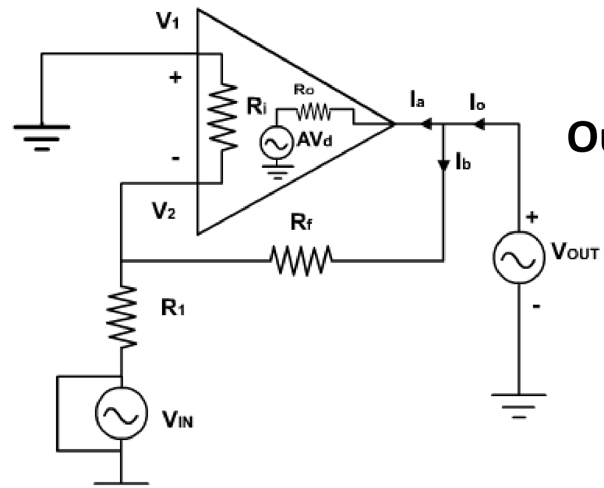
$$v_1 = v_2 = 0V$$

$$\frac{v_{in}}{R_1} = \frac{-v_o}{R_f}$$

$$v_o = -\frac{R_f}{R_1} v_{in}$$

Input Impedance:





Output Impedance:

$i_O = i_a + i_b$

Since R_O is very small

as compared to $R_f + (R_1 \,||\, R_2)$

Therefore, i.e. $i_O = i_a$

$v_O = R_O\, i_O + A\, v_d.$

$v_d = v_i - v_2 = 0 - B\, v_O$

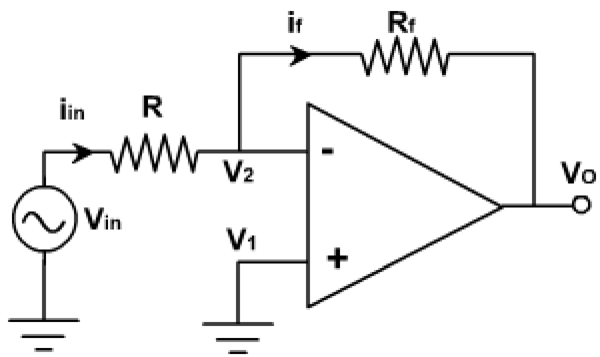
$$i_o = \frac{v_o - Av_d}{R_o}$$

$$= \frac{v_o - ABv_o}{R_o}$$

$$R_{of} = \frac{v_o}{i_o} = \frac{R_o}{1 + AB}$$

$$\text{where, } B = R_1 / R_f$$

Analog Inverter and Scale Changer:



Assuming OPAMP to be an ideal one, the differential input voltage is zero.

i.e. $v_d = 0$

Therefore, $v_1 = v_2 = 0$

Since input impedance is very high, therefore, input current is zero. OPAMP do not sink any current.

$$i_{in} = i_f$$

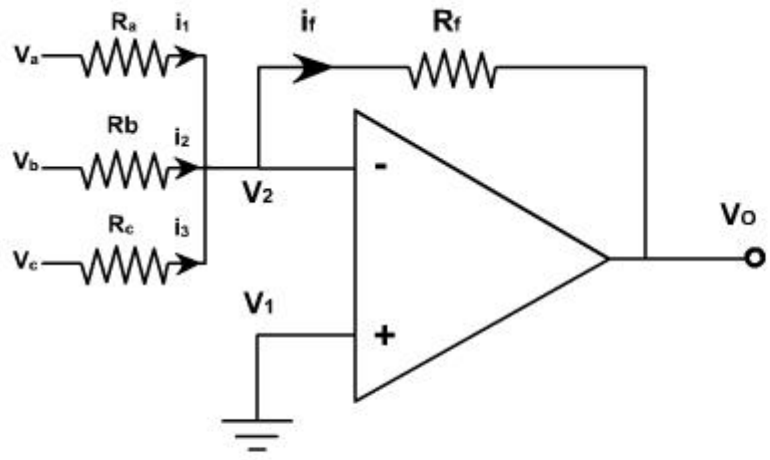
$$v_{in} / R = - v_O / R_f$$

$$v_o = - (R_f / R)\, v_{in}$$

If $R = R_f$ then $v_O = -v_{in}$, the circuit behaves like an inverter.

If $R_f / R = K$ (a constant) then the circuit is called inverting amplifier or scale changer voltages.

Inverting summer:



for an ideal OPAMP, $v_1 = v_2$. The current drawn by OPAMP is zero. Thus, applying KCL at v_2 node.

$$i_1 + i_2 + i_3 = i_f$$

$$\frac{V_a}{R_a} + \frac{V_b}{R_b} + \frac{V_c}{R_c} = -\frac{V_o}{R_f}$$

$$V_o = -\left(\frac{R_f}{R_a}V_a + \frac{R_f}{R_b}V_b + \frac{R_f}{R_c}v_c\right)$$

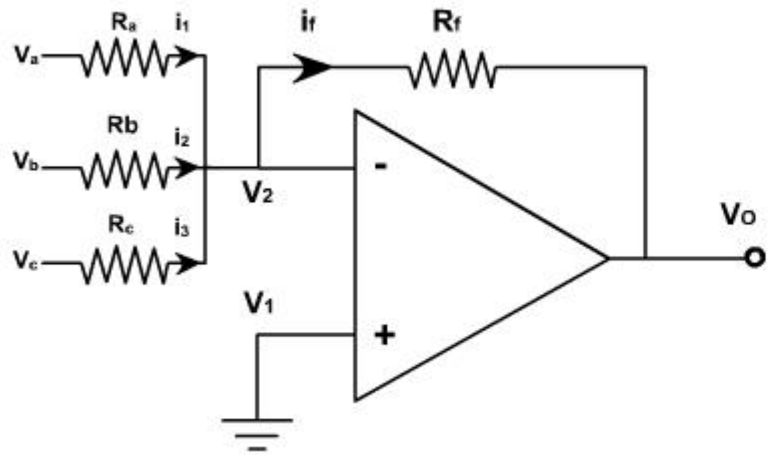
If in the circuit shown, $R_a = R_b = R_c = R$

$$V_o = -\frac{R_f}{R}(V_a + V_b + V_c)$$

This means that the output voltage is equal to the negative sum of all the inputs times the gain of the circuit R_f/R; hence the circuit is called a summing amplifier. When $R_f = R$ then the output voltage is equal to the negative sum of all inputs.

$$v_o = -(v_a + v_b + v_c)$$

Inverting summer:



If each input voltage is amplified by a different factor in other words weighted differently at the output, the circuit is called then scaling amplifier.

$$\frac{R_f}{R_a} \neq \frac{R_f}{R_b} \neq \frac{R_f}{R_c}$$

$$V_o = -\left(\frac{R_f}{R_a}V_a + \frac{R_f}{R_b}V_b + \frac{R_f}{R_c}v_c\right)$$

The circuit can be used as an averaging circuit, in which the output voltage is equal to the average of all the input voltages.

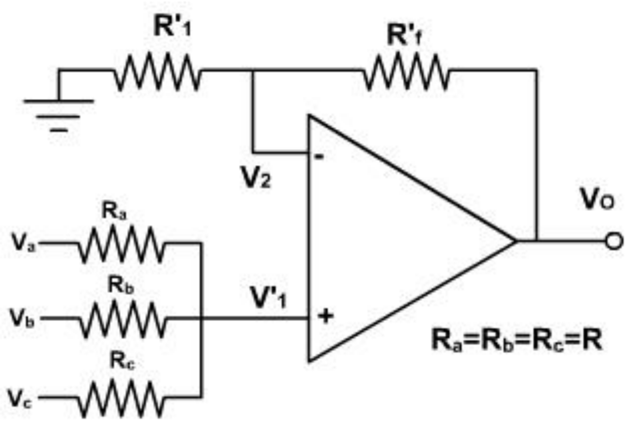
In this case, Ra= Rb= Rc = R and Rf / R = 1 / n where n is the number of inputs.

Here Rf / R = 1 / 3.

vo = -(va+ vb + vc) / 3

In all these applications input could be either ac or dc.

Non-inverting summer:



If the input voltages are connected to non-inverting input through resistors, then the circuit can be used as a summing or averaging amplifier through proper selection of R1, R2, R3 and Rf. as shown.

To find the output voltage expression, v1 is required. Applying superposition theorem, the voltage v1 at the non-inverting terminal is given by

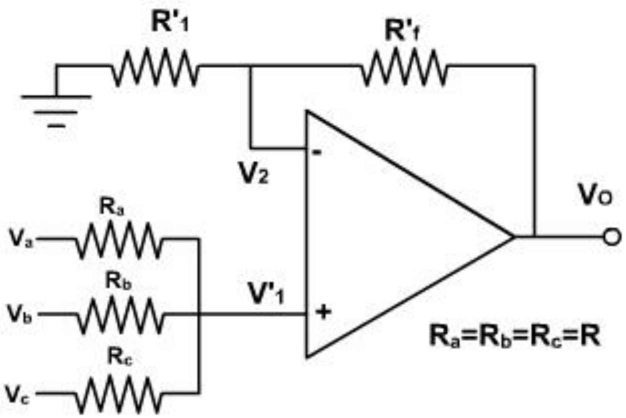
$$v_1 = \frac{R/2}{R+R/2} v_a + \frac{R/2}{R+R/2} v_b + \frac{R/2}{R+R/2} v_c$$

$$V_1 = \frac{V_a + V_b + V_c}{3}$$

Hence the output voltage is:

$$V_o = \left(1+\frac{R_f}{R_1}\right) V_1 = \left(1+\frac{R_f}{R_1}\right)\left(\frac{V_a + V_b + V_c}{3}\right)$$

Non-inverting summer:



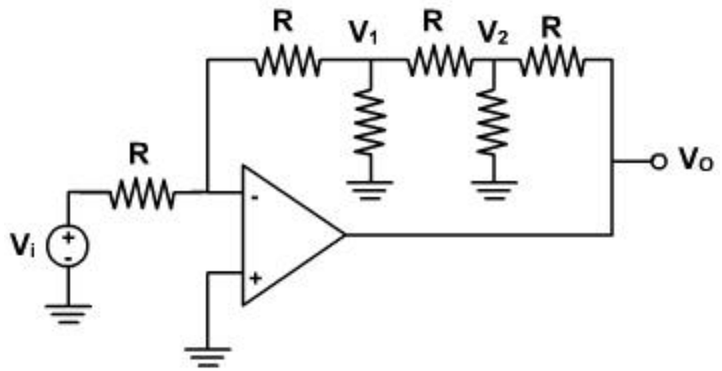
This shows that the output is equal to the average of all input voltages times the gain of the circuit (1+ R_f / R_1), hence the name averaging amplifier.

If (1+R_f/ R_1) is made equal to 3 then the output voltage becomes sum of all three input voltages.

$$v_o = v_a + v_b + v_c$$

Hence, the circuit is called summing amplifier.

Q. Find the gain of VO / Vi of the circuit.



Sol:

Current entering at the inveting terminal $= \dfrac{V_i}{R}$ $\quad\therefore\quad \dfrac{0 - V_1}{R} = \dfrac{V_i}{R} \Rightarrow V_1 = -V_i$

Applying KCL to node 1,

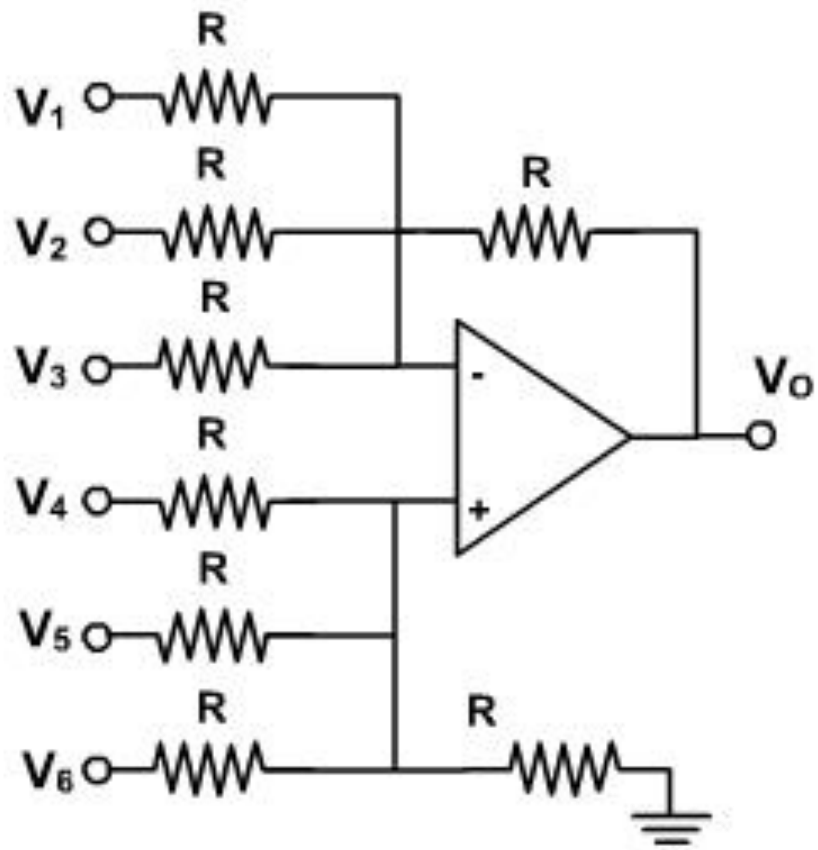
$$\frac{0 - V_1}{R} = \frac{V_1}{R} + \frac{V_1 - V_2}{R}$$
$$V_2 = -3V$$

Applying KCL to node 2,

$$\frac{V_1 - V_2}{R} = \frac{V_2}{R} + \frac{V_2 - V_0}{R}$$
$$8V_i = -V_0$$
$$\text{or,}\quad \frac{V_0}{V_i} = -8$$

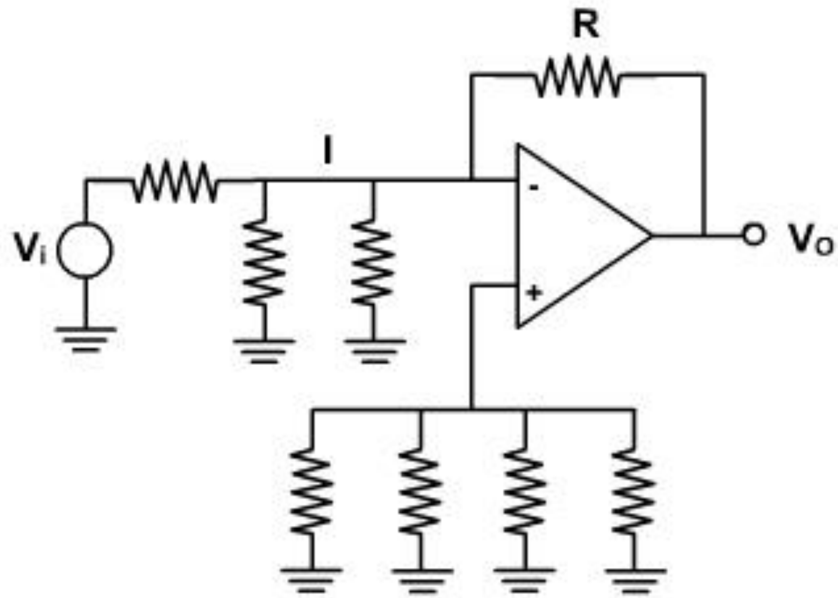
Thus the gain A = -8 V / V_O

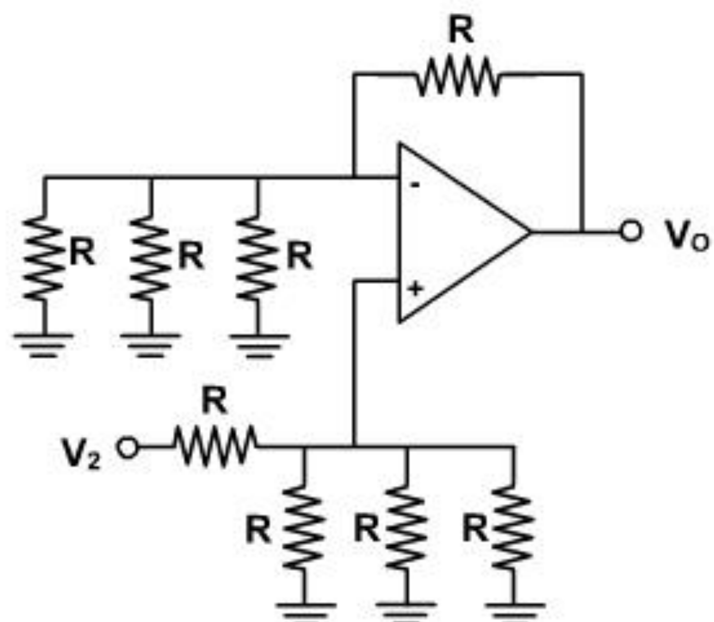
Q. Find a relationship between V_O and V_1 through V_6 in the circuit .



Sol:

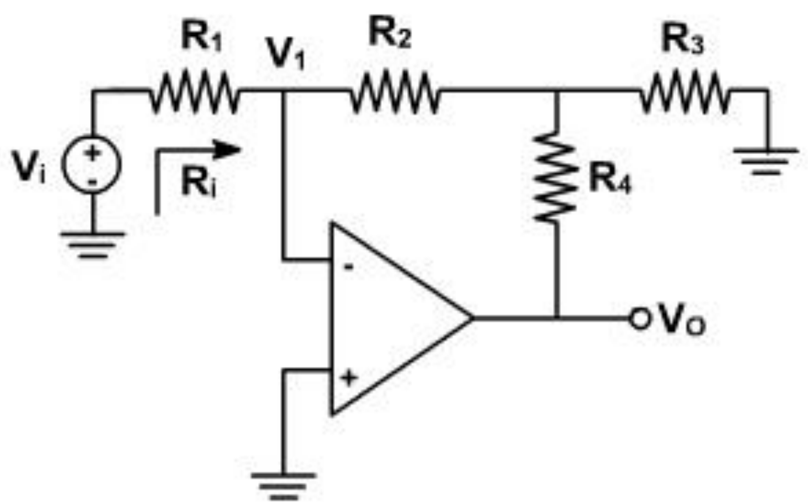
$V_O = V_2 + V_4 + V_6 - V_1 - V_3 - V_5.$



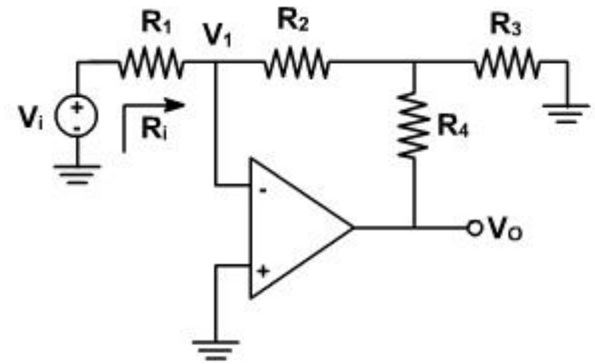


Q. (a) Show that the circuit of given figure has $A = V_O / V_i = -K (R_2 / R_1)$ with $K = 1 + R_4 / R_2 + R_4 / R_3$, and $R_i = R_1$.

(b) Specify resistance not larger than 100 K to achieve A = -200 V / V and R_i = 100 K.



Sol:



$$\frac{V_i - V_1}{R_1} = \frac{V_1 - V_2}{R_2} = \frac{V_2 - V_0}{R_4} + \frac{V_2 - 0}{R_3}$$

$$\Rightarrow \quad \frac{V_i}{R_1} = \frac{V_2}{R_2} = \frac{V_2 - V_0}{R_4} + \frac{V_2}{R_3}$$

$$\therefore \quad V_2 = -V_i \frac{R_2}{R_1}$$

$$\frac{V_i}{R_1} = \frac{V_2 - V_0}{R_4} + \frac{V_2}{R_3}$$

$$\therefore \quad \frac{V_i}{R_1} = \frac{V_i \frac{R_2}{R_1} - V_0}{R_4} + \frac{V_i \frac{R_2}{R_1}}{R_3}$$

$$\therefore \quad \frac{V_i}{R_1} = \frac{V_i R_2 - V_0 R_1}{R_1 R_4} + \frac{V_i V_2}{R_1 R_3}$$

or $\quad V_i R_4 R_3 = -V_i R_2 R_3 - V_0 R_1 R_3 - V_i R_2 R_4$

or $\quad V_i(R_3 R_4 + R_2 R_3 + R_2 R_4) = V_0 R_1 R_3$

$$\therefore \quad \frac{V_0}{V_i} = \frac{R_3 R_4 + R_2 R_3 + R_2 R_4}{R_1 R_3}$$

$$= -\frac{R_2}{R_1}\left(\frac{R_4}{R_2} + 1 + \frac{R_4}{R_3}\right)$$

$$= \frac{R_2}{R_1}\left(1 + R_4\left\{\frac{R_2 + R_3}{R_2 R_3}\right\}\right)$$

$$= -K\frac{R_2}{R_1} \text{ where } K = 1 + R_4\left(\frac{R_2 + R_3}{R_2 R_3}\right)$$

Given $\quad R_i = 100K$

$\therefore \quad R_1 = 100K$

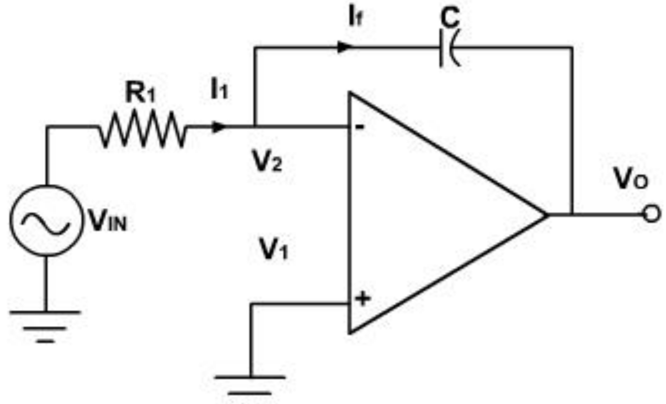
assuming $\quad R_2 = R_4 = 100K$

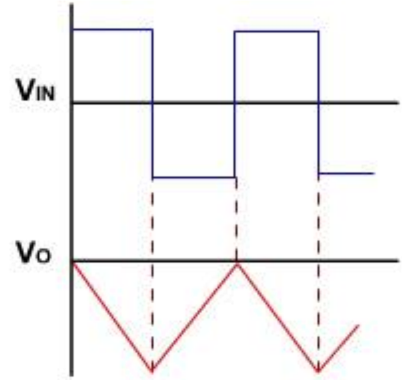
$$\left|\frac{V_0}{V_i}\right| = \frac{R_2}{R_1}\left(1 + R_4\left\{\frac{R_2 + R_3}{R_2 + R_3}\right\}\right) = 200$$

or $\quad 1 + 100K\left(\frac{100K + R_3}{100K + R_3}\right) = 200$

$\therefore \quad R3 = 505\Omega$

Integrator:





$$\frac{v_{in} - 0}{R} = C\,\frac{d(0 - v_o)}{dt}$$

$$\int_0^t \frac{v_{in}}{R}\,dt = \int_0^t C\,\frac{d(-v_o)}{dt}\,dt$$

$$= C(-v_o) + v_o\Big|_{t=0}$$

if $V_O\big|_{t=0} = 0$ V, then

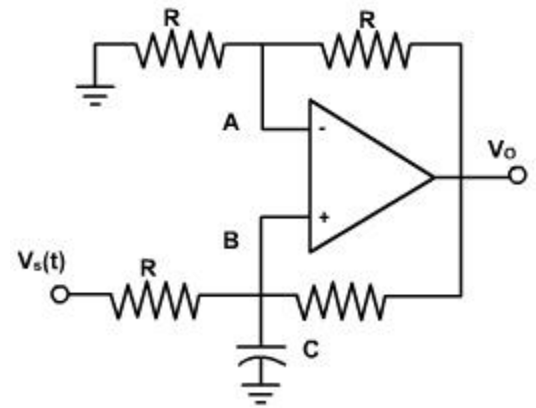
$$v_o = \frac{-1}{R}\int_0^t v_{in}\,dt$$

The output voltage is directly proportional to the negative integral of the input voltage and inversely proportional to the time constant RC.

If the input is a square wave, the output will be a triangular wave. For accurate integration, the time period of the input signal T must be longer than or equal to RC.

Q. Prove that the network shown in figure is a non-inverting integrator with

$$v_o = \frac{2}{RC}\int v_s(t)dt$$



The voltage at point A is v_O / 2 and it is also the voltage at point B because different input voltage is negligible.

$$v_B = V_O / 2$$

Therefore, applying Node current equation at point B,

$$\frac{v_B - v_A}{R} + \frac{v_B - v_o}{R} + C\frac{dv_B}{dt} = 0$$

or

$$\frac{2v_B}{R} - \frac{v_s}{R} - \frac{v_o}{R} + C\frac{dv_B}{dt} = 0$$

or

$$\frac{v_o}{R} - \frac{v_s}{R} - \frac{v_o}{R} + \frac{C}{2}\left(\frac{dv_o}{dt}\right) = 0$$

$$\therefore \frac{dv_o}{dt} = \frac{2v_s}{RC}$$

and

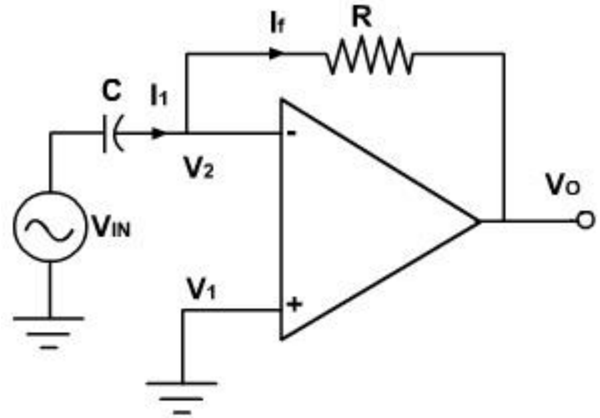
$$v_o = \int \frac{2}{RC} v_s$$

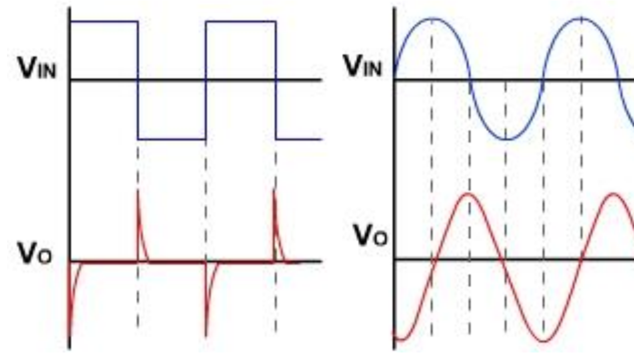
Differentiator:

Since, $i_{in} = i_f$

Therefore, $C\dfrac{d}{dt}(V_{in} - 0) = \dfrac{0 - V_o}{R}$

$$V_o = -RC\frac{dV_{in}}{dt}$$



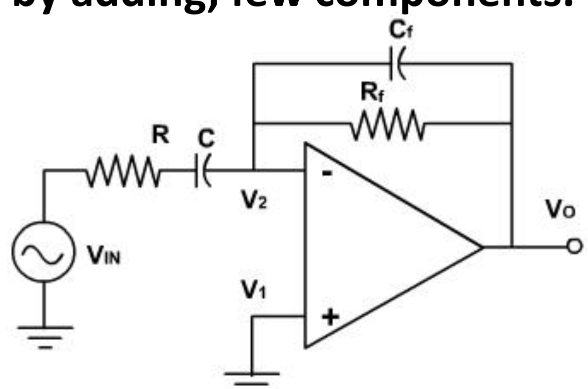


Thus the output v_o is equal to the RC times the negative instantaneous rate of change of the input voltage v_{in} with time. A sine wave input produces cosine output.

The input signal will be differentiated properly if the time period T of the input signal is larger than or equal to R_f C.

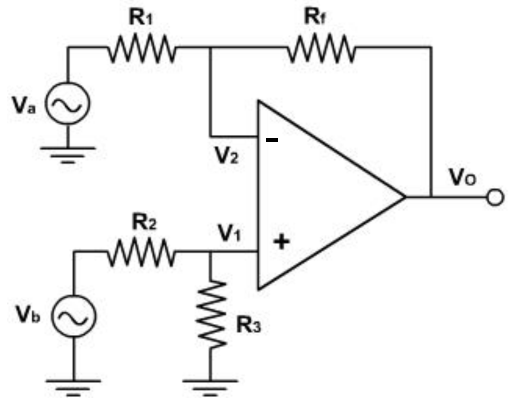
$$T \geq R_f C$$

As the frequency changes, the gain changes. Also at higher frequencies the circuit is highly susceptible at high frequency noise and noise gets amplified. Both the high frequency noise and problem can be corrected by adding, few components.



Differential Amplifier:

OP-AMP Applications



Since there are two inputs superposition theorem can be used to find the output voltage. When V_b= 0, then the circuit becomes inverting amplifier, hence the output due to V_a only is

$$V_{o(a)} = -(R_f / R_1)\, V_a$$

Similarly when, V_a = 0, the configuration is a inverting amplifier having a voltage divided network at the non-inverting input

$$V_1 = \frac{R_3}{R_2 + R_3} V_b$$

The output due to v_b only is

$$V_{o(b)} = \left(1 + \frac{R_f}{R_1}\right)\left(\frac{R_3}{R_2 + R_3}\right) V_b$$

$$= \left(\frac{R_1 + R_f}{R_1}\right)\left(\frac{R_3}{R_2 + R_3}\right) V_b$$

If $R_1 = R_2$ & $R_f = R_3$ then

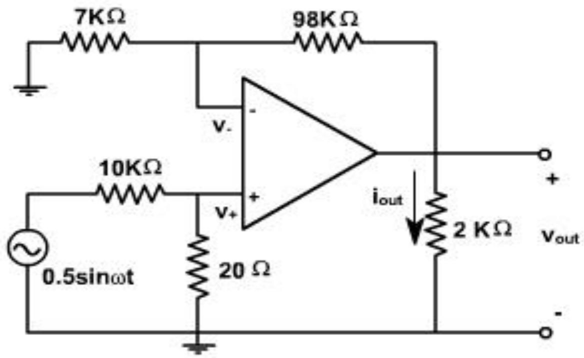
$$V_{o(b)} = \frac{R_f}{R_1} V_b$$

Therefore the, total output voltage v_o is given by

$$V_o = V_{o(a)} + V_{o(b)}$$

$$V_o = \frac{R_f}{R_1}(-V_a + V_b)$$

Q. Find v_{out} and i_{out} for the circuit shown in figure. The input voltage is sinusoidal with amplitude of 0.5 V.



We begin by writing the KCL equations at both the + and – terminals of the op-amp. For the negative terminal,

$$\frac{v_- - v_{out}}{9.8 \times 10^4} + \frac{v_- - 0}{7000} = 0$$

Therefore,

$$15\, v_- = v_{out}$$

For the positive terminal,

$$\frac{v_+ - v_{in}}{10^4} + \frac{v_+ - 0}{2 \times 10^4} = 0$$

This yields two equations in three unknowns, v_{out}, v_+ and v_-. The third equation is the relationship between v_+ and v_- for the ideal OPAMP,

$$v_+ = v_-$$

Solving these equations, we find

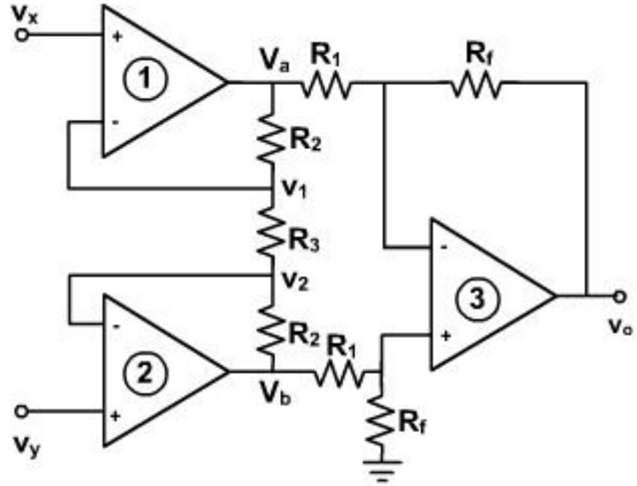
$$v_{out} = 10\, v_{in} = 5 \sin\omega t \text{ V}$$

Since 2 kΩ resistor forms the load of the op-amp, then the current i_{out} is given by

$$i_{out} = \frac{V_{out}}{R_{load}} \times 2.5 \sin\omega t \text{ mA}$$

Q. For the different amplifier shown in figure, verify that

$$v_o = -\left(1 + \frac{2R_2}{R_3}\right)\frac{R_f}{R_1}(V_x - V_y)$$



An **instrumentation amplifier** is a differential op-amp circuit providing high input impedances with ease of gain adjustment through the variation of a single resistor (R3 in the Figure if other resistances are equal).

Since the differential input voltage of OPAMP is negligible, therefore,

$$v_1 = v_x$$
$$\text{and } v_2 = v_y$$

The input impedance of OPAMP is very large and, therefore, the input current of OPAMP is negligible. Amd The OPAMP3 is working as differential amplifier, therefore,

$$\frac{v_a - v_1}{R_2} = \frac{v_1 - v_2}{R_3} \qquad (E-1)$$

$$\frac{v_1 - v_2}{R_3} = \frac{v_2 - v_b}{R_2} \qquad (E-2)$$

$$\frac{v_a - v_x}{R_2} = \frac{v_x - v_y}{R_3}$$

$$v_a = \frac{R_2}{R_3}(v_x - v_y) + v_x$$

$$\frac{v_x - v_y}{R_3} = \frac{v_y - v_b}{R_2}$$

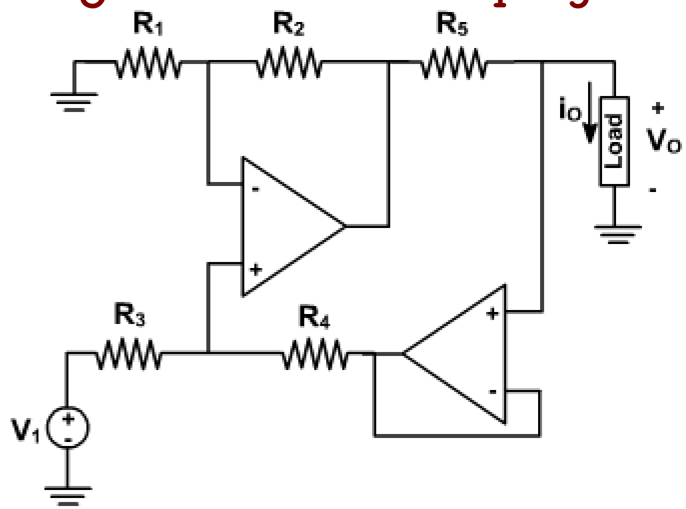
$$v_b = v_y - \frac{R_2}{R_3}(v_x - v_y)$$

$$v_o = \frac{R_f}{R_1}(v_b - v_a)$$

$$= \frac{R_f}{R_1}\left\{v_y - v_x - 2\frac{R_2}{R_3}(v_x - v_y)\right\}$$

$$v_o = -\left(1 + \frac{2R_2}{R_3}\right)\frac{R_f}{R_1}(v_x - v_y)$$

Q. Obtain an expression of the type $i_O = V_i / R - V_O / R_O$ for the circuit shown in fig. Hence verify that if $R_4 / R_3 = R_2 / R_1$ the circuit is a V-I converter with $R_O = \infty$ and $R = R_1 R_5 / R_2$.



i_O = current through the resistor.

$$\frac{R_4}{R_3} = \frac{R_2}{R_1}$$

Q. Design an amplifier with a gain of +9 and Rf =12 KΩ using an op-Amp.

Soln:

Since the gain is positive:

Choose a non-inverting amplifier

Then we have,

$$V_O = \left[1 + \frac{R_f}{R_1}\right] V_i$$

Gain is,

$$1 + \frac{R_f}{R_1} = 9$$

$$\frac{R_f}{R_1} = 8$$

$$\therefore R_1 = \frac{R_f}{8} = \frac{12 \times 10^3}{8}$$

$$R_1 = 1.5\ K\Omega$$

Q. A 5 mV peak voltage, 1 KHz signal is applied to the input of an Op-Amp integrator for which R=100KΩ and C=1μF. Find the output voltage.

Soln:

Given R=100KΩ

C=1μF

V_m =5mV

F=1KHz

V_0 =?

We have $V_i = V_m \text{Sin}wt = V_m \text{Sin}2\pi ft$

V_i=5sin2000πt mV

For an integrator,

$$V_O = -\frac{1}{RC}\int V_i dt$$

on solving,

$$V_O = \frac{1}{40\pi}\cos 2000\pi t \quad mV$$

Q. The input to a differentiator is a sinusoidal voltage of peak value 5mV and frequency 2KHz . Find the output if R = 100KΩ and C=1μF.

Soln:

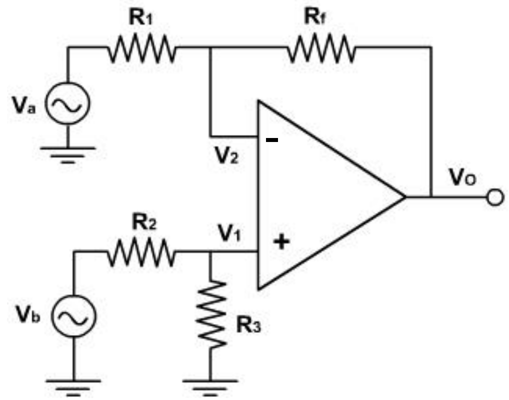
$$V_i = 5\sin 4000\pi t \quad mV$$

$$for\ differentiator\ V_O = -RC\frac{dV_i}{dt}$$

$$on\ \ solving\ V_O = -2000\pi \cos 4000\pi t \quad mV$$

:

OP-AMP Applications



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Similarly when, V_a = 0, the configuration is a inverting amplifier having a voltage divided network at the non-inverting input

$$V_1 = \frac{R_3}{R_2 + R_3} V_b$$

The output due to v_b only is

$$V_{o(b)} = \left(1 + \frac{R_f}{R_1}\right)\left(\frac{R_3}{R_2 + R_3}\right) V_b$$

$$= \left(\frac{R_1 + R_f}{R_1}\right)\left(\frac{R_3}{R_2 + R_3}\right) V_b$$

If $R_1 = R_2$ & $R_f = R_3$ then

$$V_{o(b)} = \frac{R_f}{R_1} V_b$$

Therefore the, total output voltage v_o is given by

$$V_o = V_{o(a)} + V_{o(b)}$$

$$V_o = \frac{R_f}{R_1}(-V_a + V_b)$$

-

555

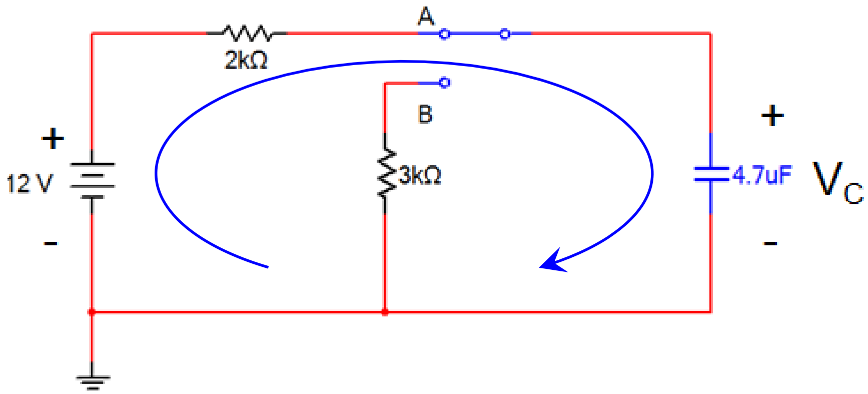
WHAT IS A 555 TIMER?

- The 555 timer is an 8-pin IC that is capable of producing accurate time delays and/or oscillators.
- In the time delay mode, the delay is controlled by one external resistor and capacitor.
- In the oscillator mode, the frequency of oscillation and duty cycle are both controlled with two external resistors and one capacitor.

CAPACITOR

- A capacitor is an electrical component that can temporarily store a charge (voltage).
- The rate that the capacitor charges/discharges is a function of the capacitor's value and its resistance.
- To understand how the capacitor is used in the 555 Timer oscillator circuit, you must understand the basic charge and discharge cycles of the capacitor.

Capacitor Charge Cycle



- Capacitor is initially discharged.
- Switch is moved to position A.
- Capacitor will charge to +12 v.
- Capacitor will charge through the 2 KΩ resistor.

<u>Equation for Charging Capacitor</u>

$$V_C = \left(V_{Final} - V_{Initial}\right) \times \left(1 - e^{-t/RC}\right) + V_{Initial}$$

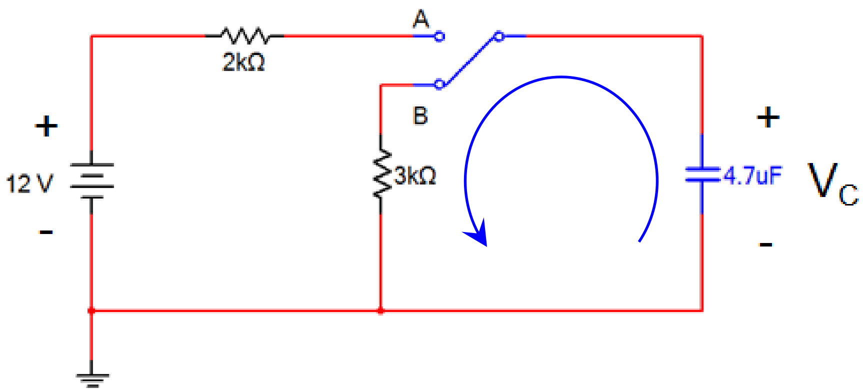
Where:

V_C = The voltage across the capacitor

V_{Final} = The voltage across the capacitor that is fully charged

$V_{Initial}$ = Any initial voltage across the capacitor as it begins to charge

Capacitor Discharge Cycle



- Capacitor is initially charged.
- Switch is moved to position B.
- Capacitor will discharge to +0 v.
- Capacitor will discharge through the 3 KΩ resistor.

Equation for Discharging Capacitor

$$V_C = (V_{Initial} - V_{Final}) \times (e^{-t/RC})$$

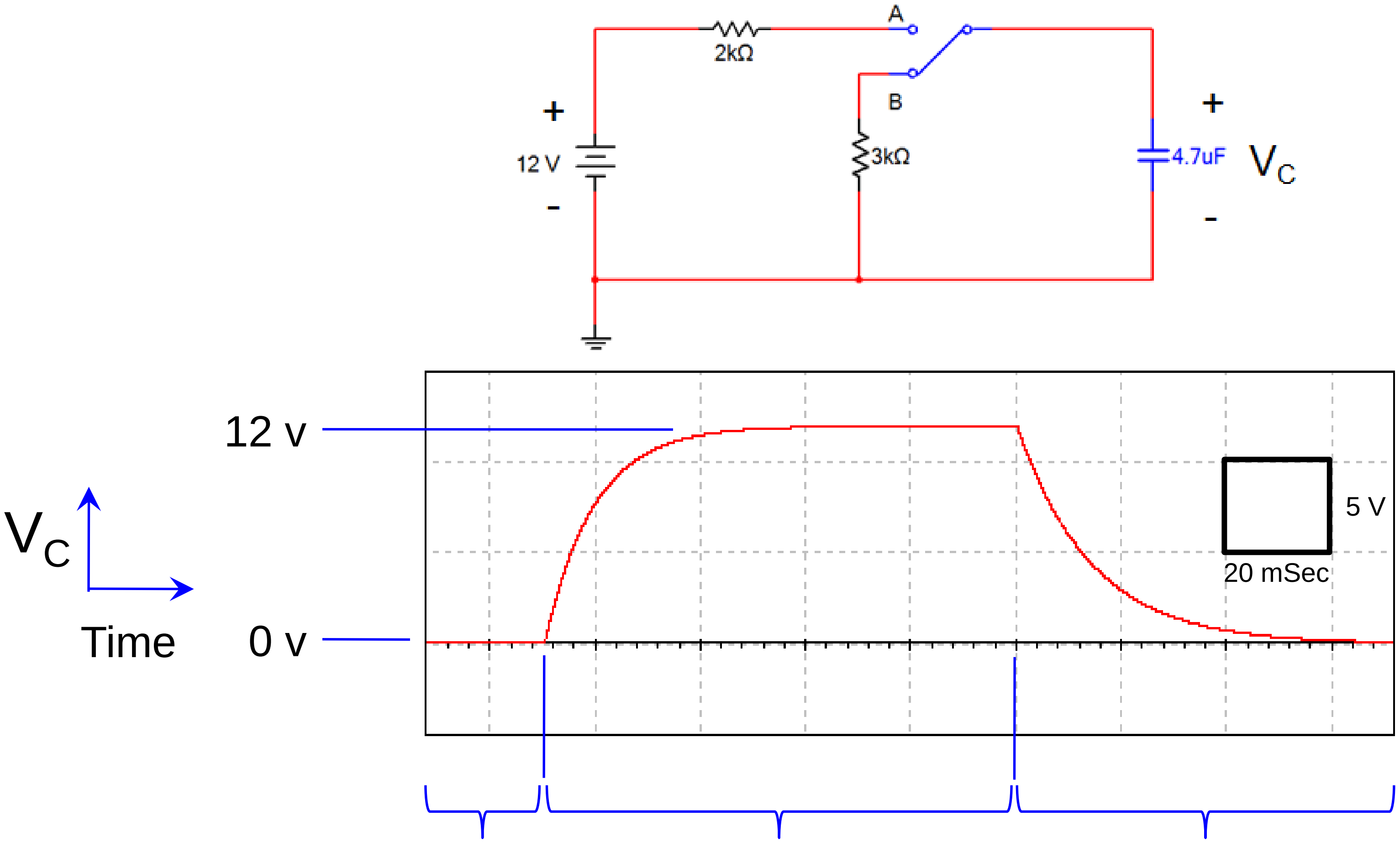
Where:

V_C = The voltage across the capacitor

V_{Final} = The voltage across the capacitor that is fully discharged

$V_{Initial}$ = Any initial voltage across the capacitor as it begins to discharge

Capacitor Charge & Discharge

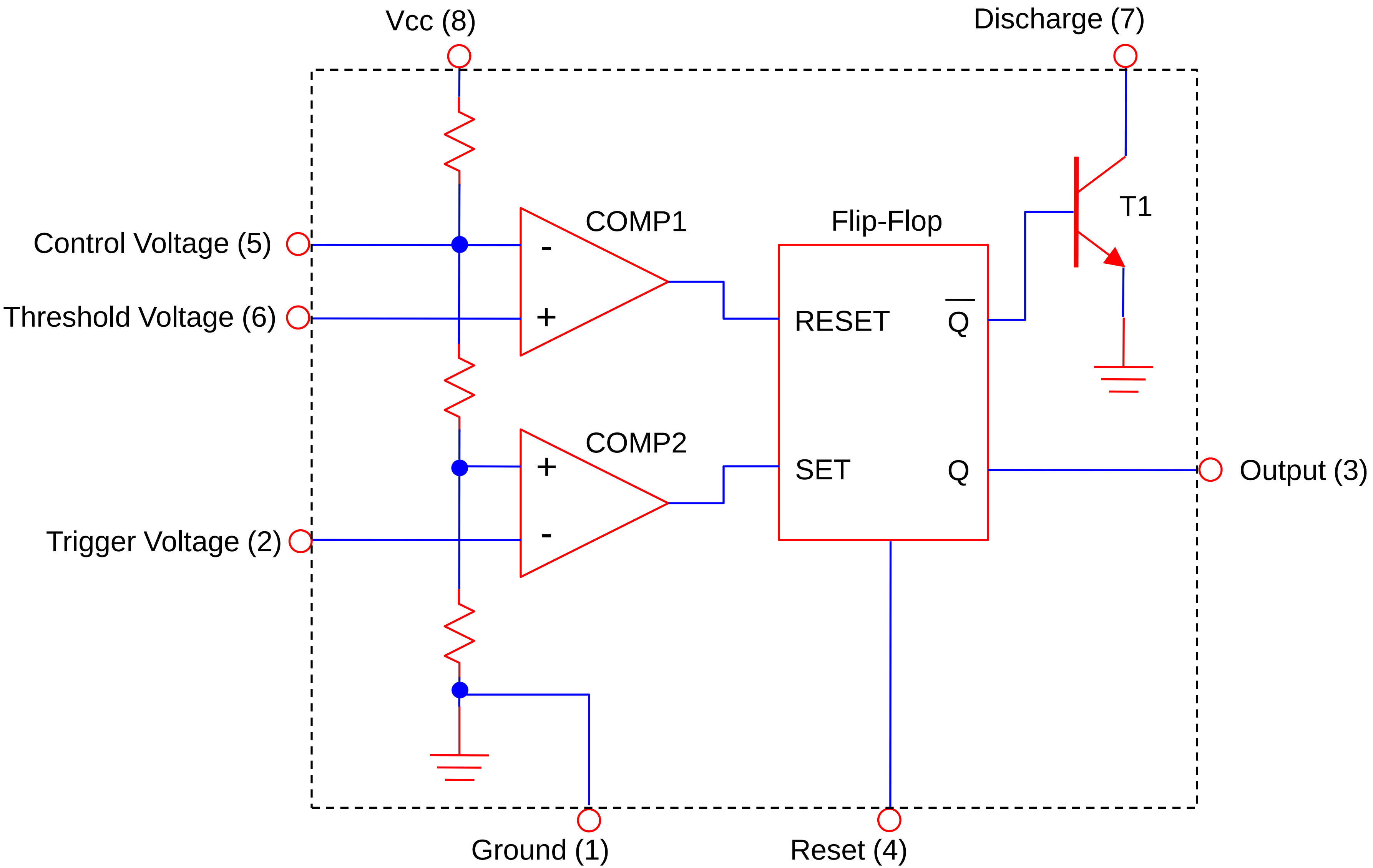


Switch has been at position B for a long period of time. The capacitor is completely discharged.

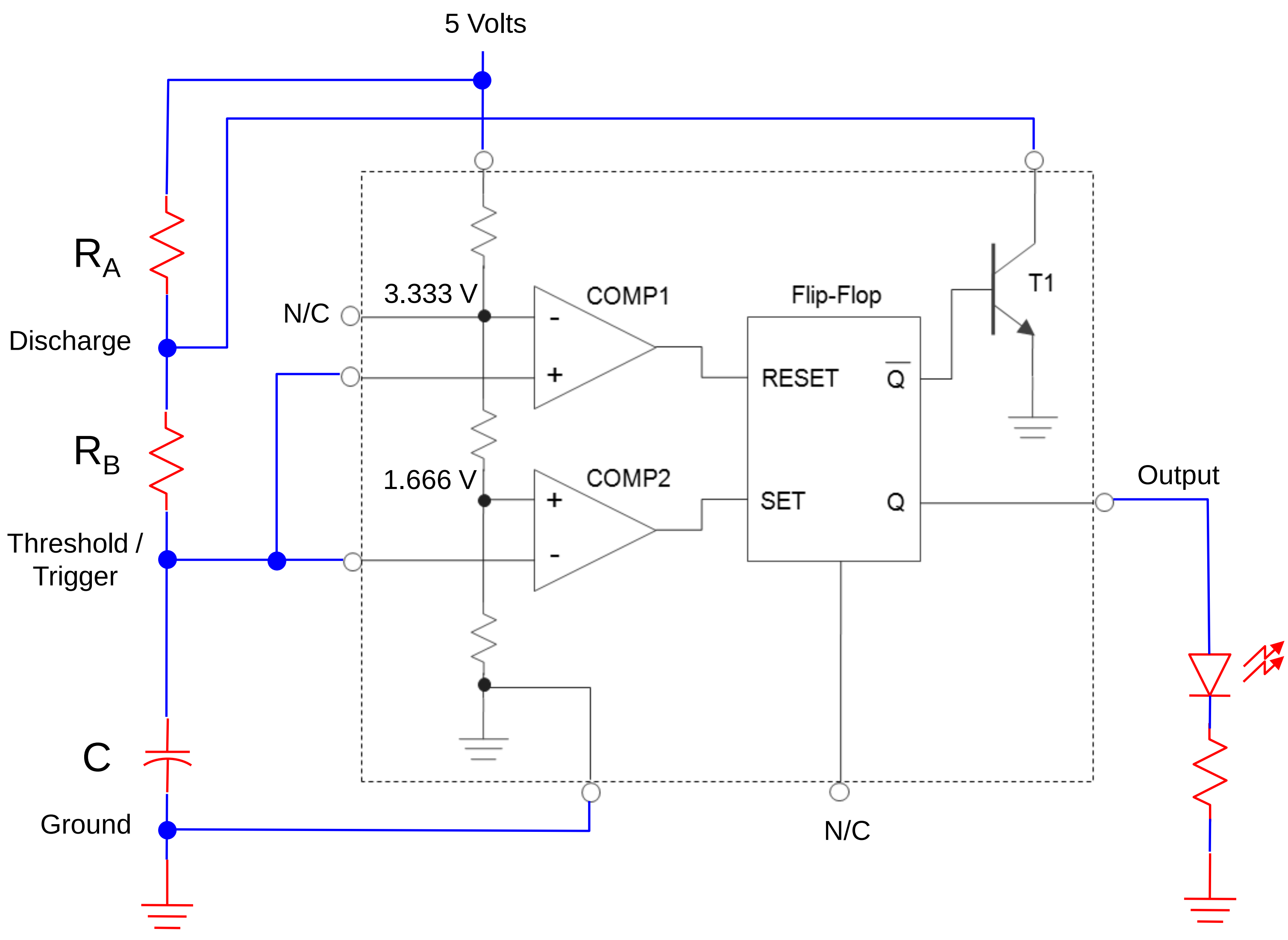
Switch is moved to position A. The capacitor charges through the 2KΩ resistor.

Switch is moved back to position B. The capacitor discharges through the 3KΩ resistors.

Block Diagram for a 555 Timer



SCHEMATIC OF A 555 TIMER IN OSCILLATOR MODE



PIN DESCRIPTION OF NE555 TIMER IC

- **Pin1 :** ground
- **Pin 2:** trigger
- **Pin 3:** output
- **Pin 4:** reset: the timer can be reset by applying a negative pulse to this pin. When not used, it is connected to $+V_{CC}$.
- **Pin 5:** Control voltage: This is used to change the threshold voltage as well as trigger voltage. When not used, the control pin should be bypassed to ground with 0.01uf capacitor to prevent any noise problem.
- **Pin 6:** threshold this is the non inverting terminal of the comparator 1.
- **Pin 7:** discharge: this pin is connected internally to the collector of the transistor. When the output is high, transistor is OFF and acts as a open circuit. When the output is low. the transistor will be in saturation and acts as short circuit.
- **Pin 8:** $+V_{CC}$ supply is connected w.r.t ground.

Pin Description of NE555 Timer IC

- Voltages at V_{TH} and V_{TL} are $2/3V_{CC}$ and $1/3V_{CC}$ respectively.
- Comparators produce 0 and 1 for $V_- > V_+$ and $V_- < V_+$ respectively.

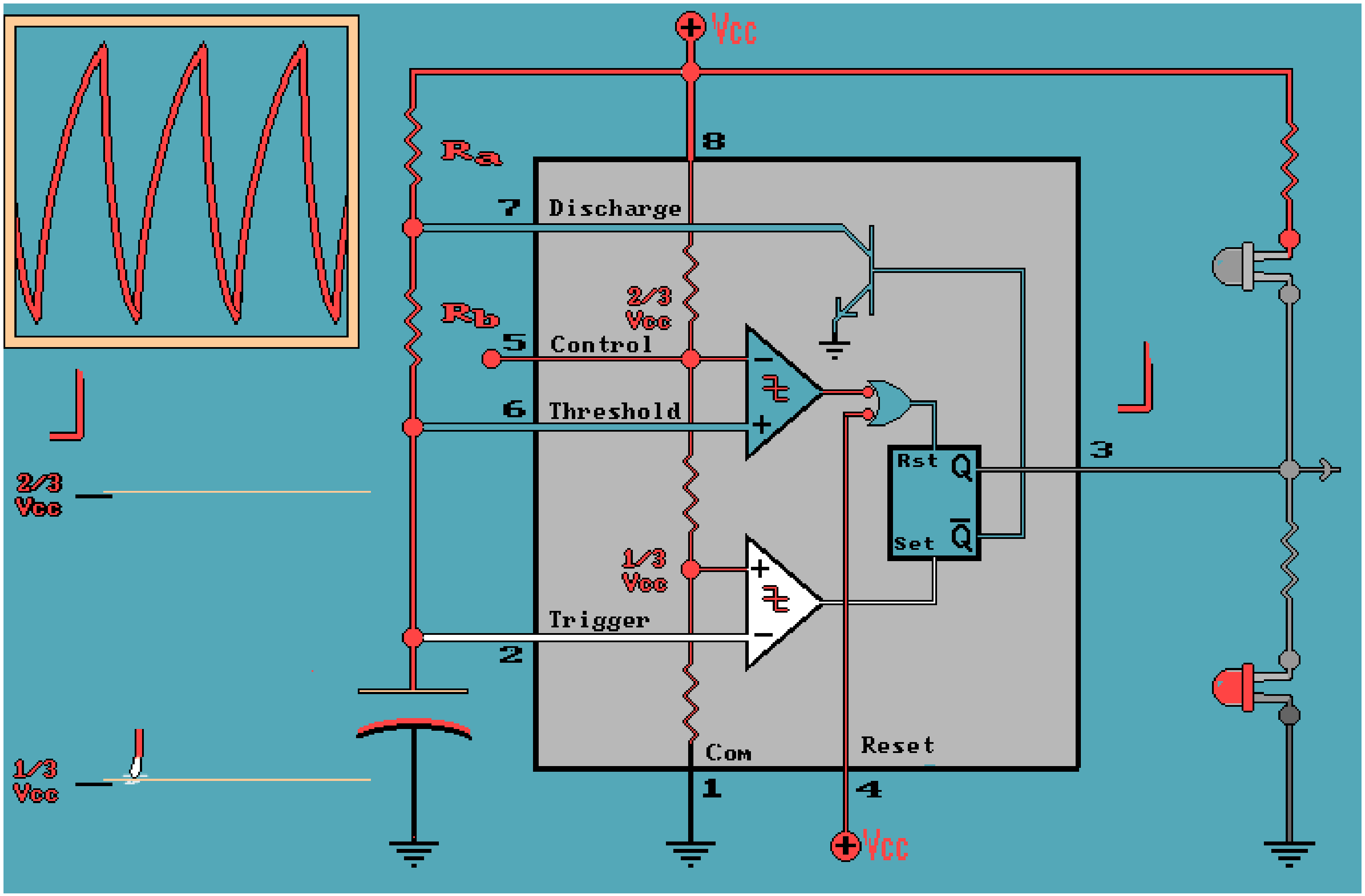
- SR latch works on the following principle

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	NA

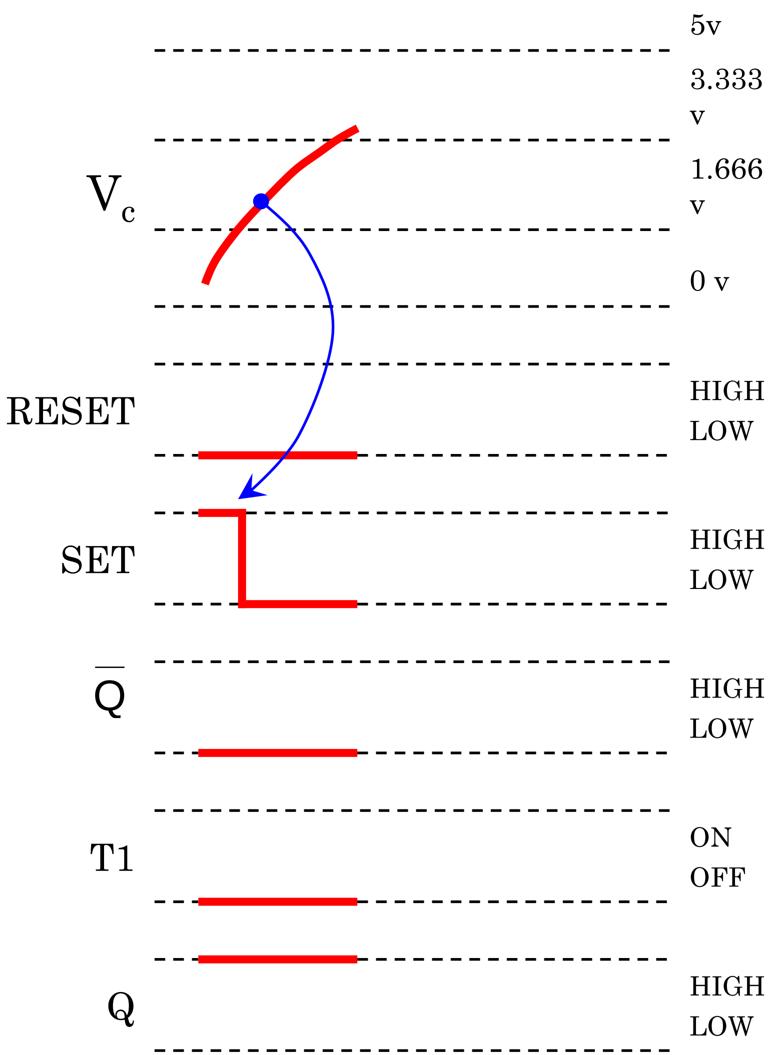
- Output is available at 3
- External triggers is generally applied at 2 no. pin of IC

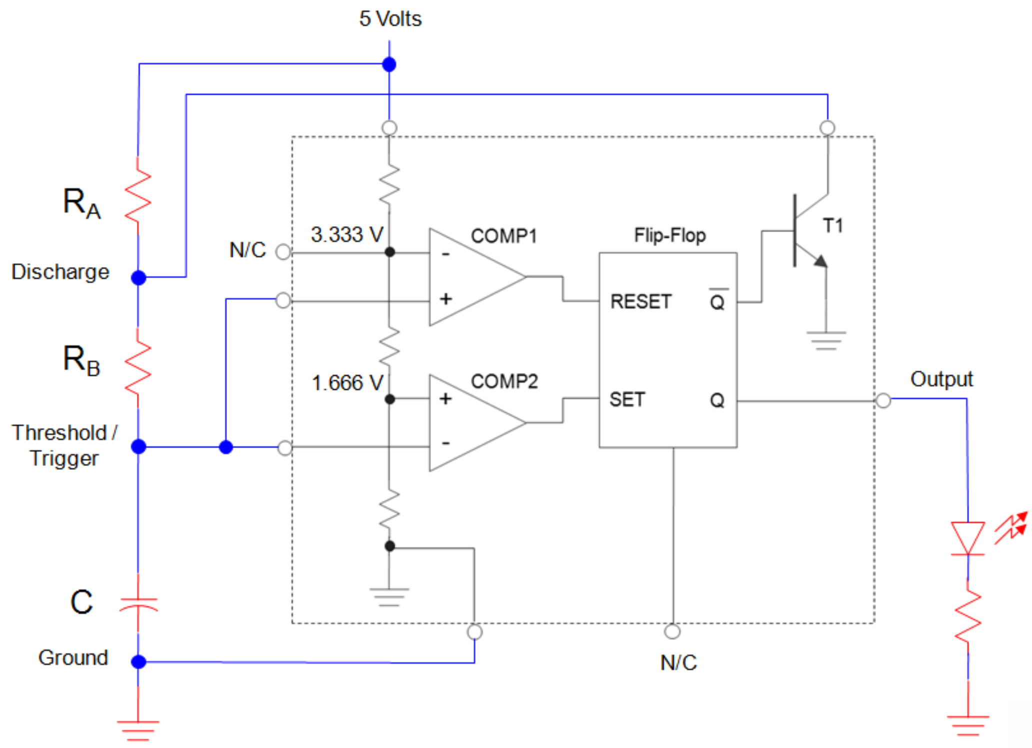
555 Oscillator
Detail Analysis

http://www.williamson-labs.com/pu-aa-555-timer_very-slow.htm

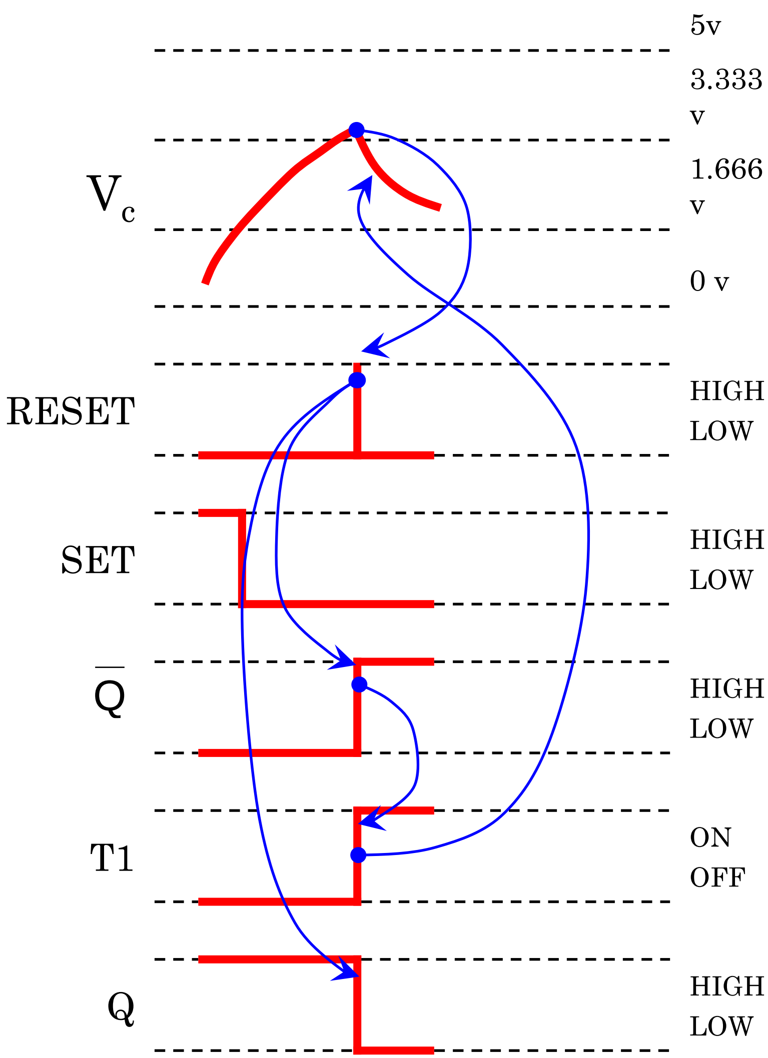


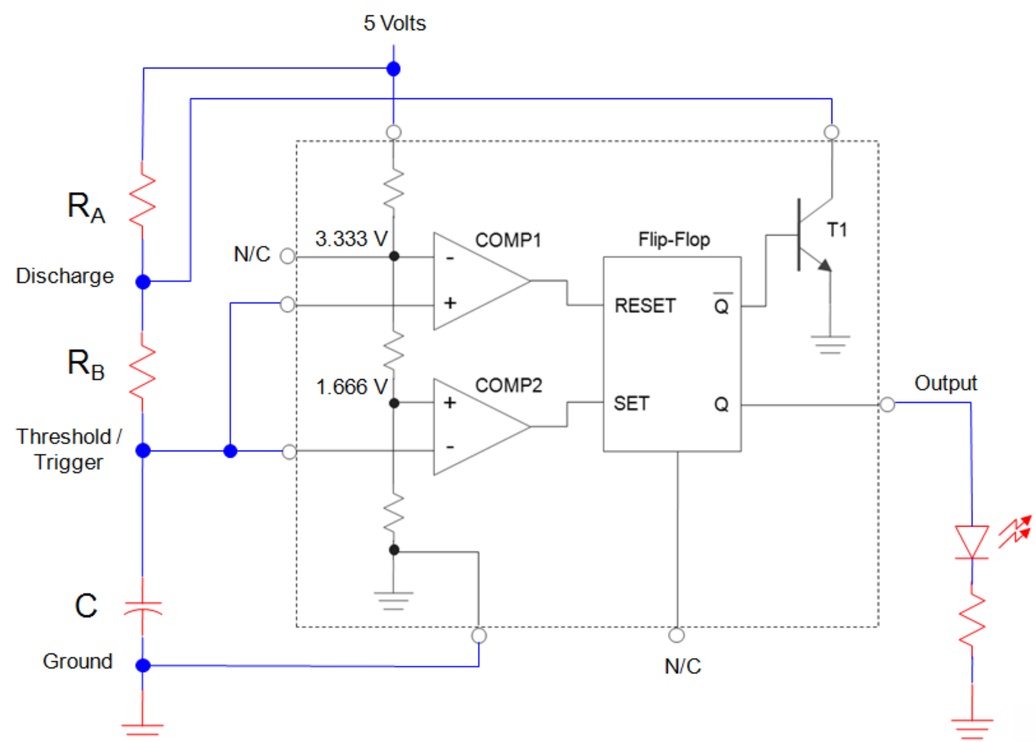
DETAIL ANALYSIS OF A 555 OSCILLATOR



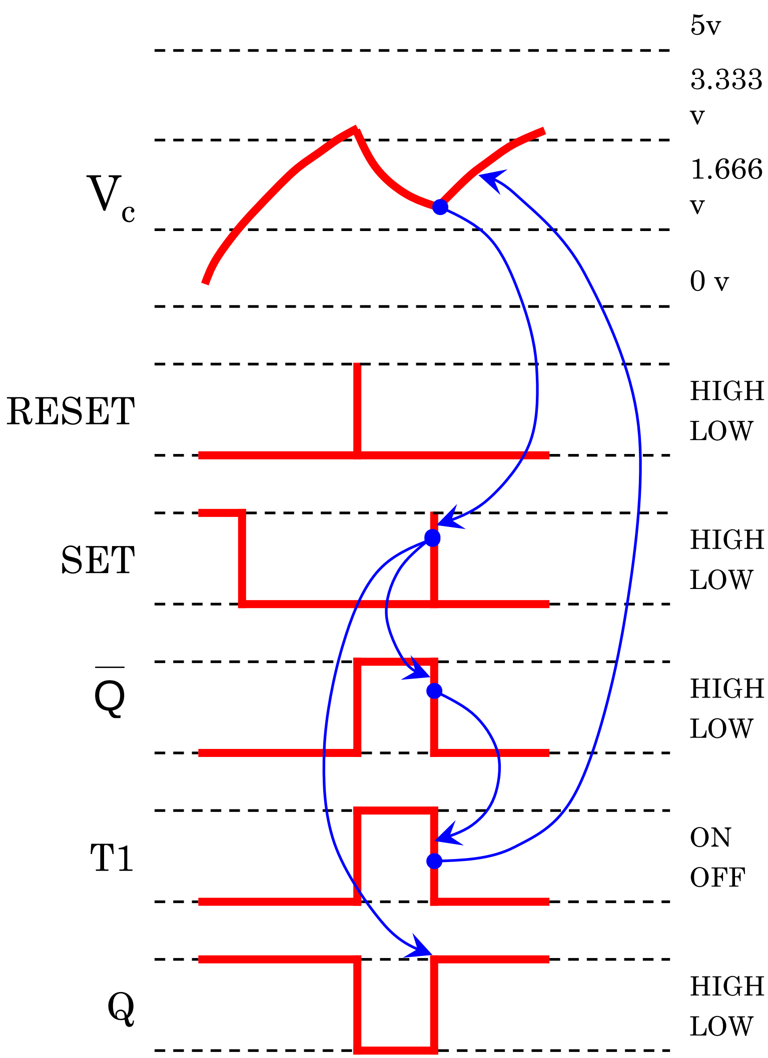


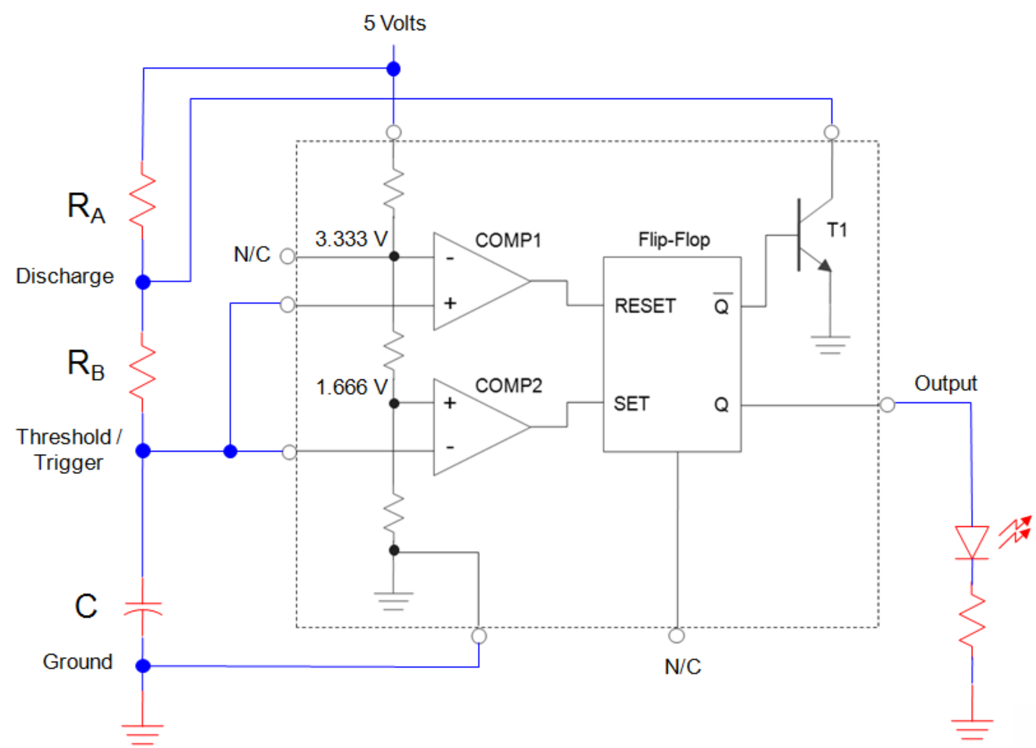
Detail Analysis of a 555 Oscillator (Conti…)





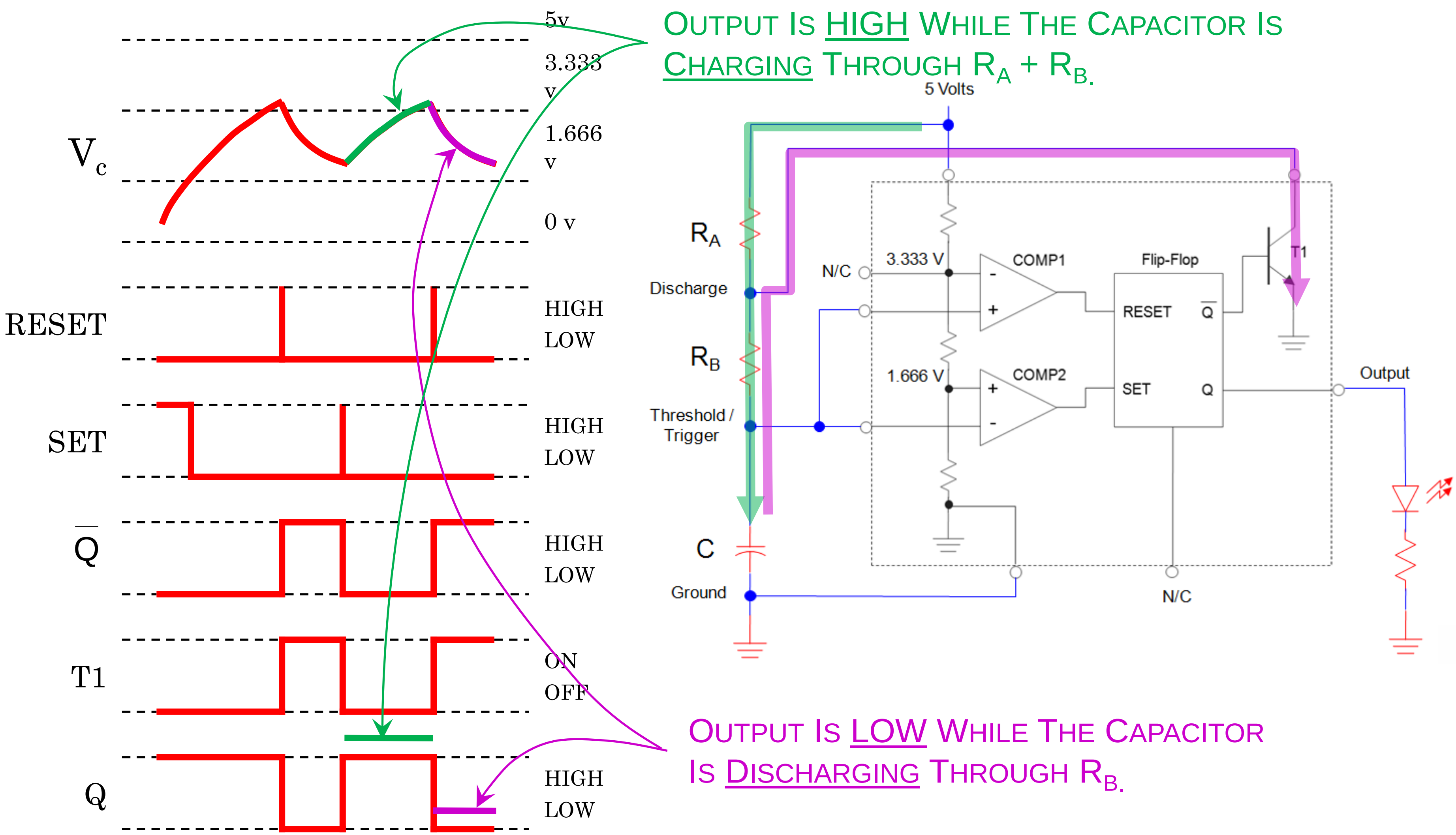
Detail Analysis of a 555 Oscillator (Conti…)





Detail Analysis of a 555 Oscillator (Conti…)

Output Is HIGH While The Capacitor Is Charging Through $R_A + R_B$.



Output Is LOW While The Capacitor Is Discharging Through R_B.

Detail Analysis of a 555 Oscillator (Conti…)

t_{HIGH} : Calculations for the Oscillator's HIGH Time

$$V_C = (V_{Final} - V_{Initial}) \times \left(1 - e^{-\frac{t}{RC}}\right) + V_{Initial}$$

$$\tfrac{2}{3} V_{CC} = (V_{CC} - \tfrac{1}{3} V_{CC}) \times \left(1 - e^{-\frac{t}{RC}}\right) + \tfrac{1}{3} V_{CC}$$

$$\tfrac{2}{3} V_{CC} = (\tfrac{2}{3} V_{CC}) \times \left(1 - e^{-\frac{t}{RC}}\right) + \tfrac{1}{3} V_{CC}$$

$$\frac{\tfrac{2}{3} V_{CC} - \tfrac{1}{3} V_{CC}}{\tfrac{2}{3} V_{CC}} = \left(1 - e^{-\frac{t}{RC}}\right)$$

$$\tfrac{1}{2} = \left(1 - e^{-\frac{t}{RC}}\right)$$

$$\tfrac{1}{2} = \left(1 - e^{-\frac{t}{RC}}\right)$$

$$-\tfrac{1}{2} = -e^{-\frac{t}{RC}}$$

$$\ln(\tfrac{1}{2}) = \ln\left(e^{-\frac{t}{RC}}\right)$$

$$-0.693 = -\frac{t}{RC}$$

$$t_{HIGH} = 0.693\, R\, C$$

$$t_{HIGH} = 0.693 (R_A + R_B) C$$

Detail Analysis of a 555 Oscillator (Conti…)

t_{LOW}: Calculations for the Oscillator's LOW Time

$$V_C = (V_{Initial} - V_{Final}) \times \left(e^{-\frac{t}{RC}}\right)$$

$$\tfrac{1}{3}V_{CC} = (\tfrac{2}{3}V_{CC} - 0) \times \left(e^{-\frac{t}{RC}}\right)$$

$$\tfrac{1}{3}V_{CC} = (\tfrac{2}{3}V_{CC}) \times \left(e^{-\frac{t}{RC}}\right)$$

$$\frac{\tfrac{1}{3}V_{CC}}{\tfrac{2}{3}V_{CC}} = \left(e^{-\frac{t}{RC}}\right)$$

$$\tfrac{1}{2} = \left(e^{-\frac{t}{RC}}\right)$$

$$\tfrac{1}{2} = \left(e^{-\frac{t}{RC}}\right)$$

$$\ln(\tfrac{1}{2}) = \ln\left(e^{-\frac{t}{RC}}\right)$$

$$-0.693 = -\frac{t}{RC}$$

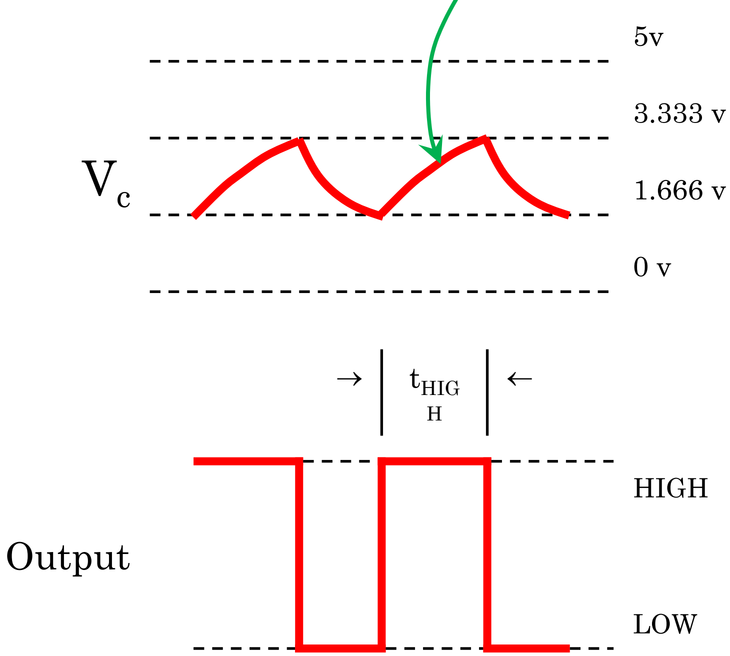
$$t_{LOW} = 0.693\,R$$

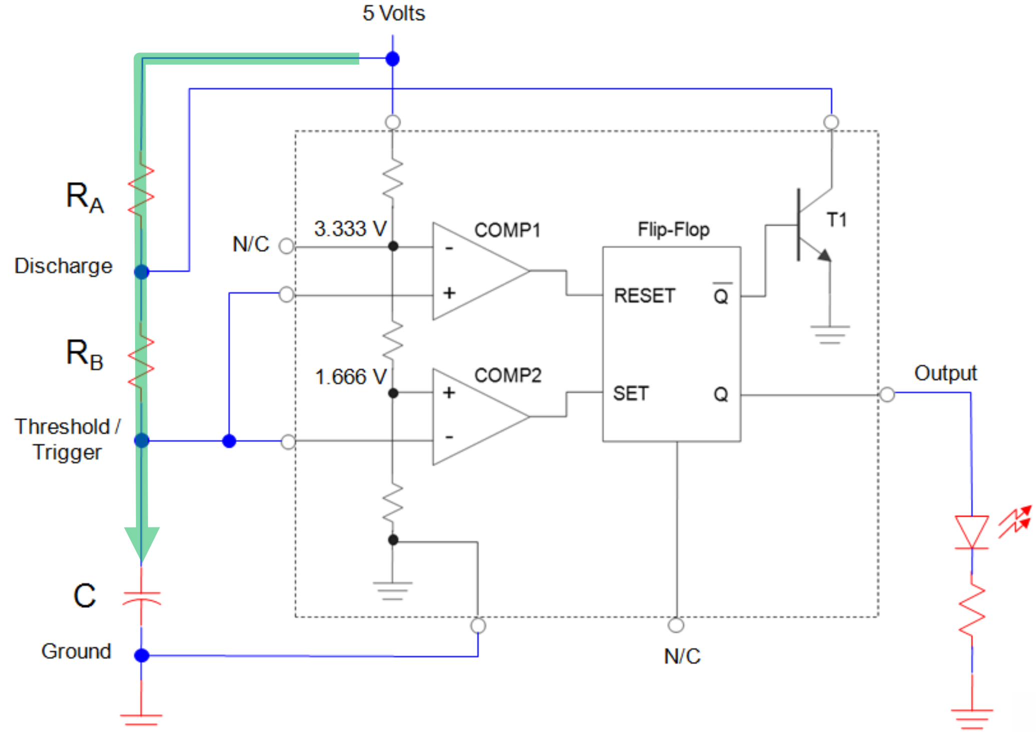
$$t_{LOW} = 0.693\,R_B C$$

555 Timer Design Equations

t_{HIGH} : Calculations for the Oscillator's HIGH Time

The Output Is HIGH While The Capacitor Is Charging Through $R_A + R_B$.



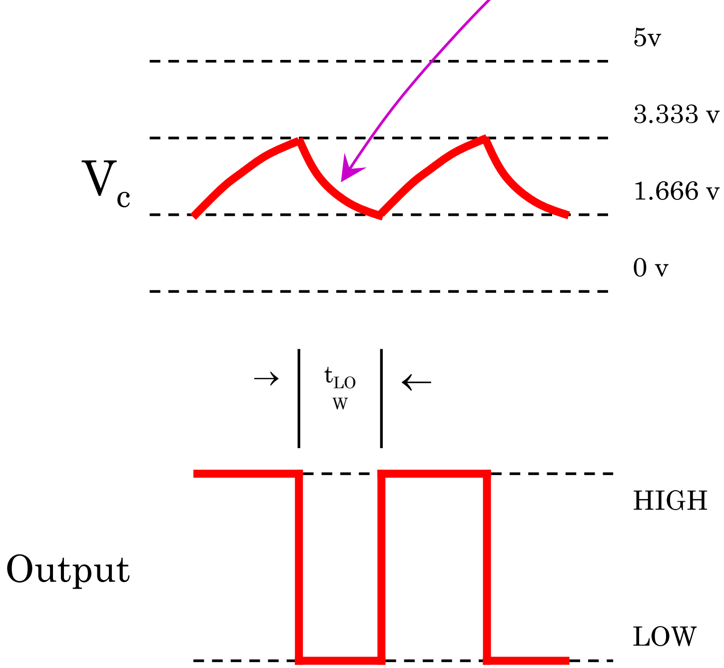


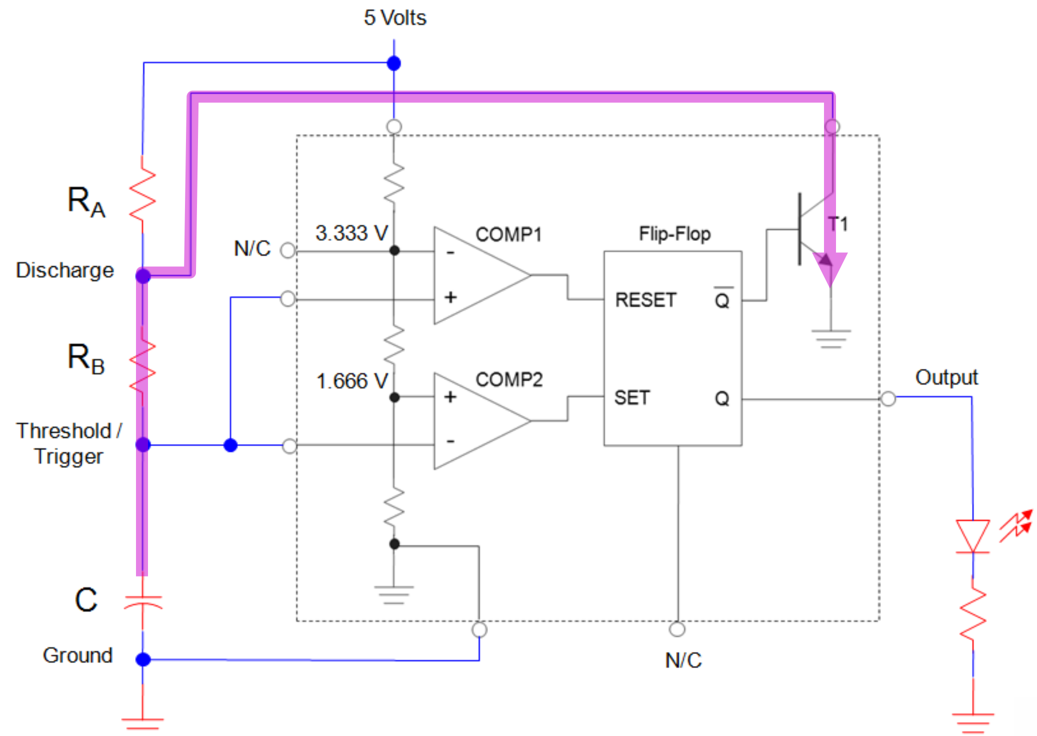
$$t_{HIGH} = 0.693(R_A + R_B)C$$

555 Timer Design Equations (Conti….)

t_{LOW} : Calculations for the Oscillator's LOW Time

The Output Is LOW While The Capacitor Is Discharging Through R_B.





$$t_{LOW} = 0.693 R_B C$$

555 Timer – Period / Frequency / DC

Period:

$$t_{HIGH} = 0.693\,(R_A + R_B)\,C$$

$$t_{LOW} = 0.693\,R_B C$$

$$T = t_{HIGH} + t_{LOW}$$

$$T = \left[0.693\,(R_A + R_B)\,C\right] + \left[0.693\,R_B C\right]$$

$$T = 0.693\,(R_A + 2R_B)\,C$$

Duty Cycle:

$$DC = \frac{t_{HIGH}}{T} \times 100\%$$

$$DC = \frac{0.693\,(R_A + R_B)\,C}{0.693\,(R_A + 2R_B)\,C} \times 100\%$$

$$DC = \frac{(R_A + R_B)}{(R_A + 2R_B)} \times 100\%$$

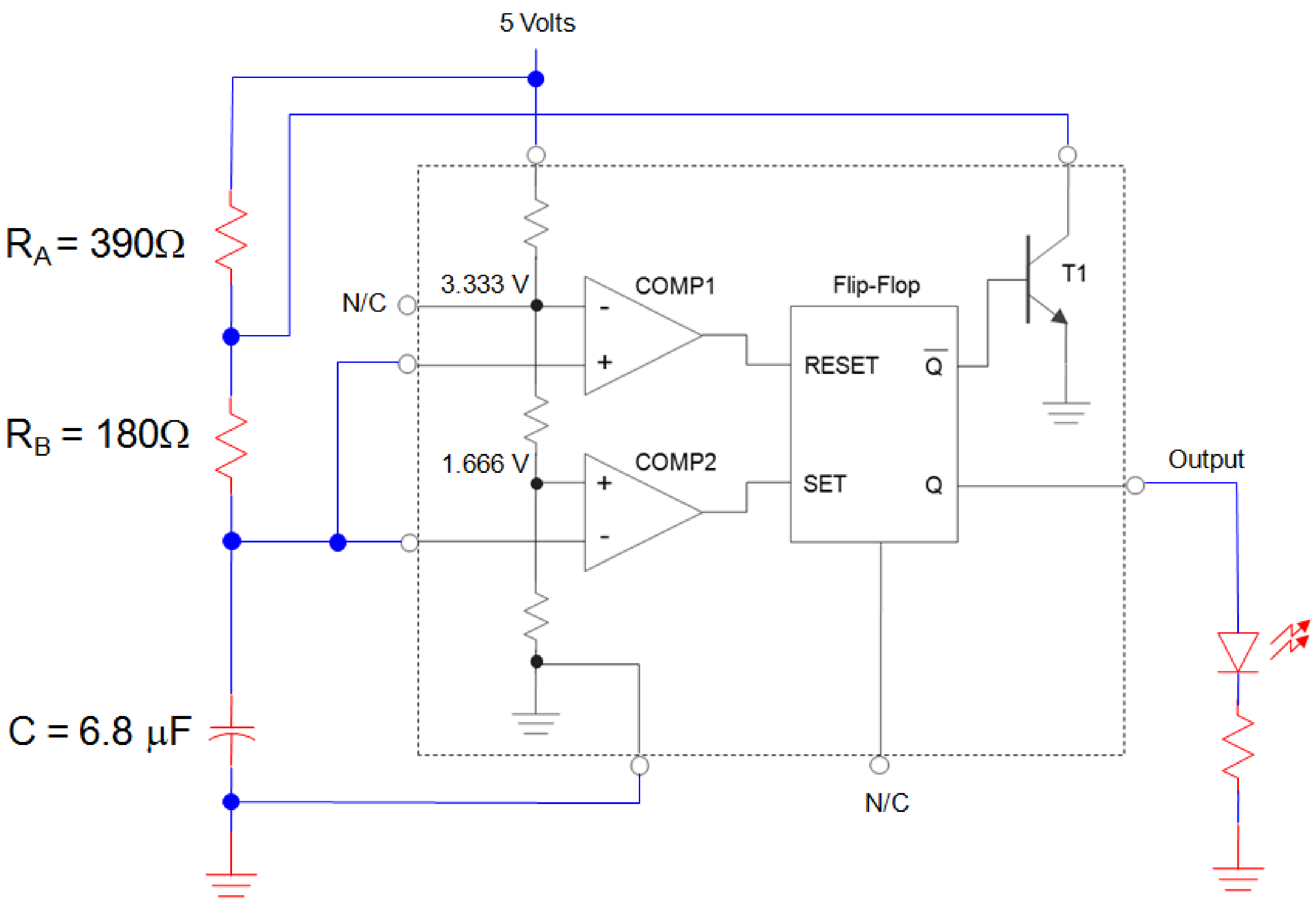
Frequency:

$$F = \frac{1}{T}$$

$$F = \frac{1}{0.693\,(R_A + 2R_B)\,C}$$

EXAMPLE: 555 OSCILLATOR

For the 555 Timer oscillator shown below, calculate the circuit's, period (T), frequency (F), and duty cycle (DC).



Example: 555 Oscillator

Solution:

$$R_A = 390\ \Omega \qquad R_B = 180\ \Omega \qquad C = 6.8\ \mu F$$

Period:

$$T = 0.693\left(R_A + 2R_B\right)C$$

$$T = 0.693\left(390\Omega + 2 \times 180\Omega\right) \times 6.8\mu F$$

$$\boxed{T = 3.534\,\text{mSec}}$$

Frequency:

$$F = \frac{1}{T}$$

$$F = \frac{1}{3.534\,\text{mSec}}$$

$$\boxed{F = 282.941\text{Hz}}$$

Duty Cycle:

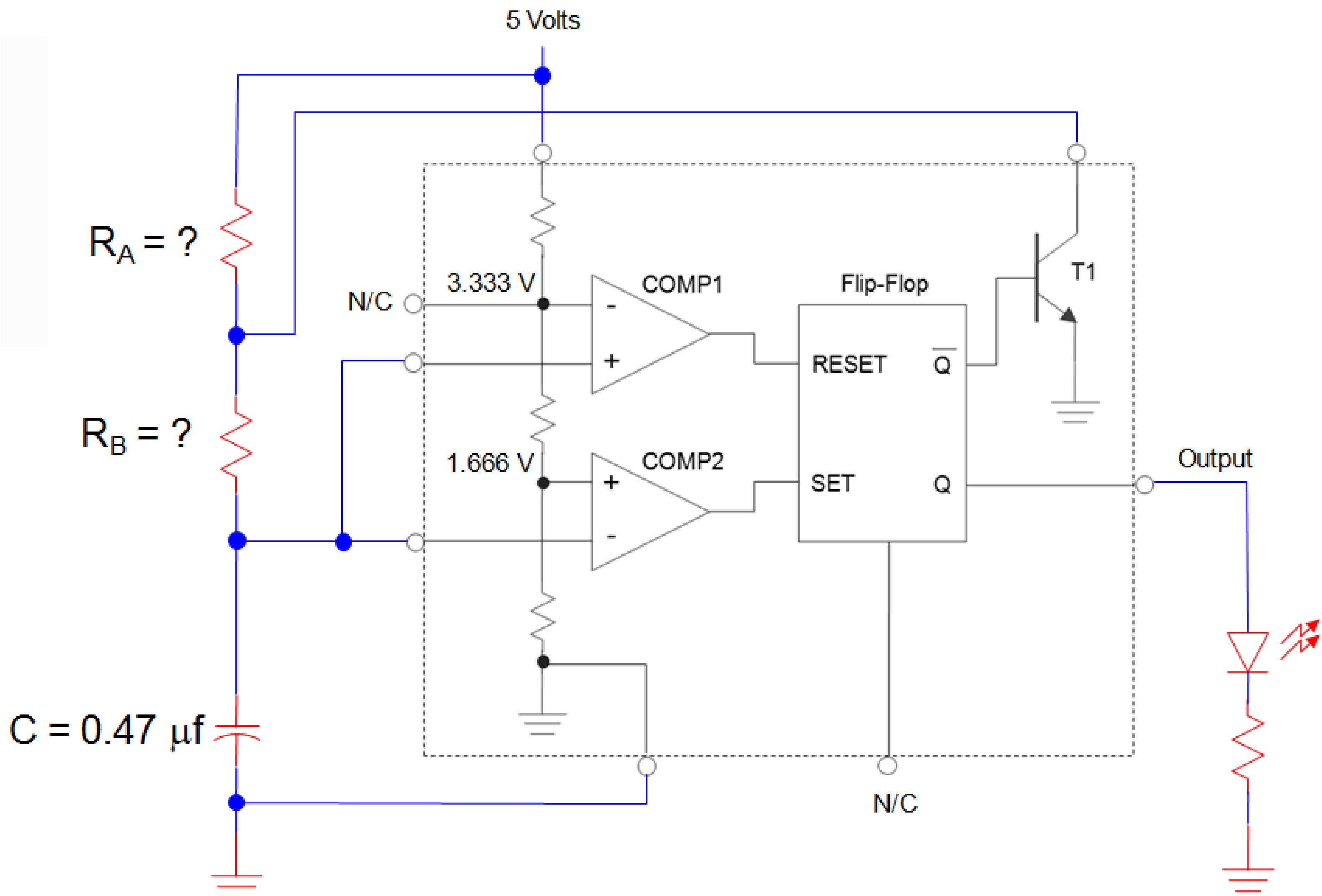
$$DC = \frac{\left(R_A + R_B\right)}{\left(R_A + 2R_B\right)} \times 100\%$$

$$DC = \frac{\left(390\,\Omega + 180\,\Omega\right)}{\left(390\,\Omega + 2 \times 180\,\Omega\right)} \times 100\%$$

$$\boxed{DC = 76\%}$$

EXAMPLE: 555 OSCILLATOR

For the 555 Timer oscillator shown below, calculate the value for R_A & R_B so that the oscillator has a frequency of 2.5 KHz @ 60% duty cycle.



Example: 555 Oscillator

Solution:

Frequency:

$$T = \frac{1}{f} = \frac{1}{2.5\,\text{kHz}} = 400\,\mu\text{Sec}$$

$$T = 0.693\,(R_A + 2R_B)\,C = 400\,\mu\text{Sec}$$

$$T = 0.693\,(R_A + 2R_B)\,0.47\,\mu\text{f} = 400\,\mu\text{Sec}$$

$$R_A + 2\,R_B = \frac{400\,\mu\text{Sec}}{0.693 \times 0.47\,\mu\text{f}} = 1228.09\,\Omega$$

$$R_A + 2\,R_B = 1228.09$$

Duty Cycle:

$$DC = \frac{(R_A + R_B)}{(R_A + 2R_B)} \times 100\% = 60\%$$

$$\frac{(R_A + R_B)}{(R_A + 2R_B)} = 0.6$$

$$R_A + R_B = 0.6(R_A + 2R_B)$$

$$R_A + R_B = 0.6 \times R_A + 1.2 \times R_B$$

$$0.4 \times R_A = 0.2 \times R_B$$

$$R_A = 0.5 \times R_B$$

Two Equations & Two Unknowns!

Example: 555 Oscillator

Solution:

Frequency:

Duty Cycle:

$$R_A + 2R_B = 1228.09$$

$$R_A = 0.5 \times R_B$$

Substitute and Solve for R_B

$$R_A + 2R_B = 1228.09\ \Omega$$

$$0.5 \times R_B + 2R_B = 1228.09\ \Omega$$

$$2.5R_B = 1228.09\ \Omega$$

$$\boxed{R_B = 491.23\ \Omega}$$

Substitute and Solve for R_A

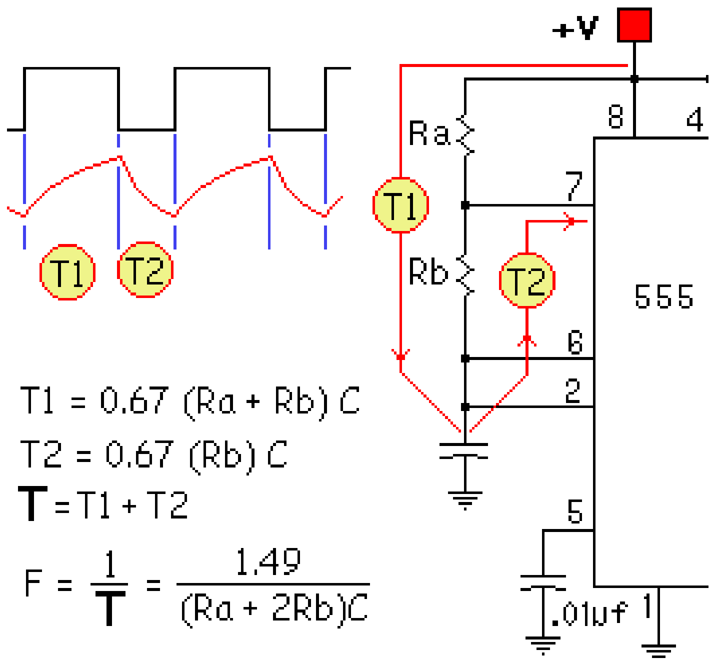
$$R_A + 2R_B = 1228.09\ \Omega$$

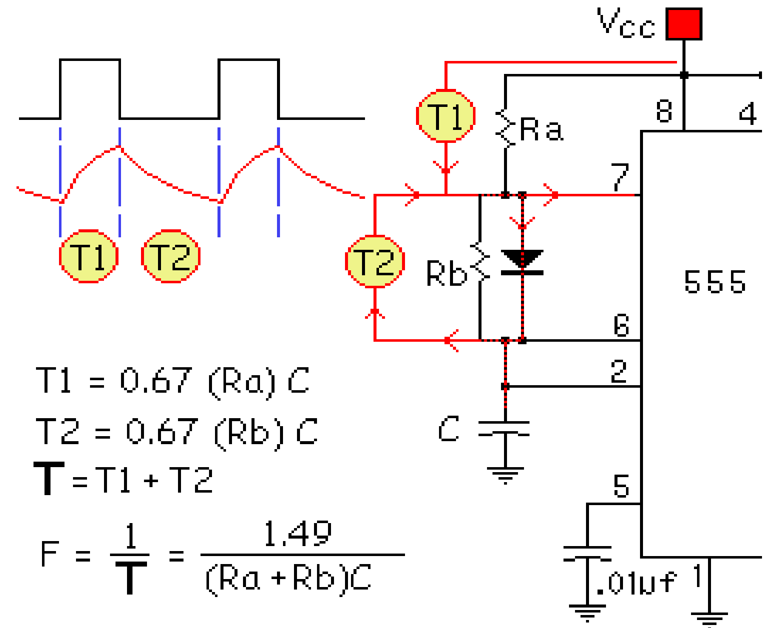
$$R_A + 2(491.23\ \Omega) = 1228.09\ \Omega$$

$$R_A + 982.472\ \Omega = 1228.09\ \Omega$$

$$\boxed{R_A = 245.618\ \Omega}$$

Astable Duty Cycle





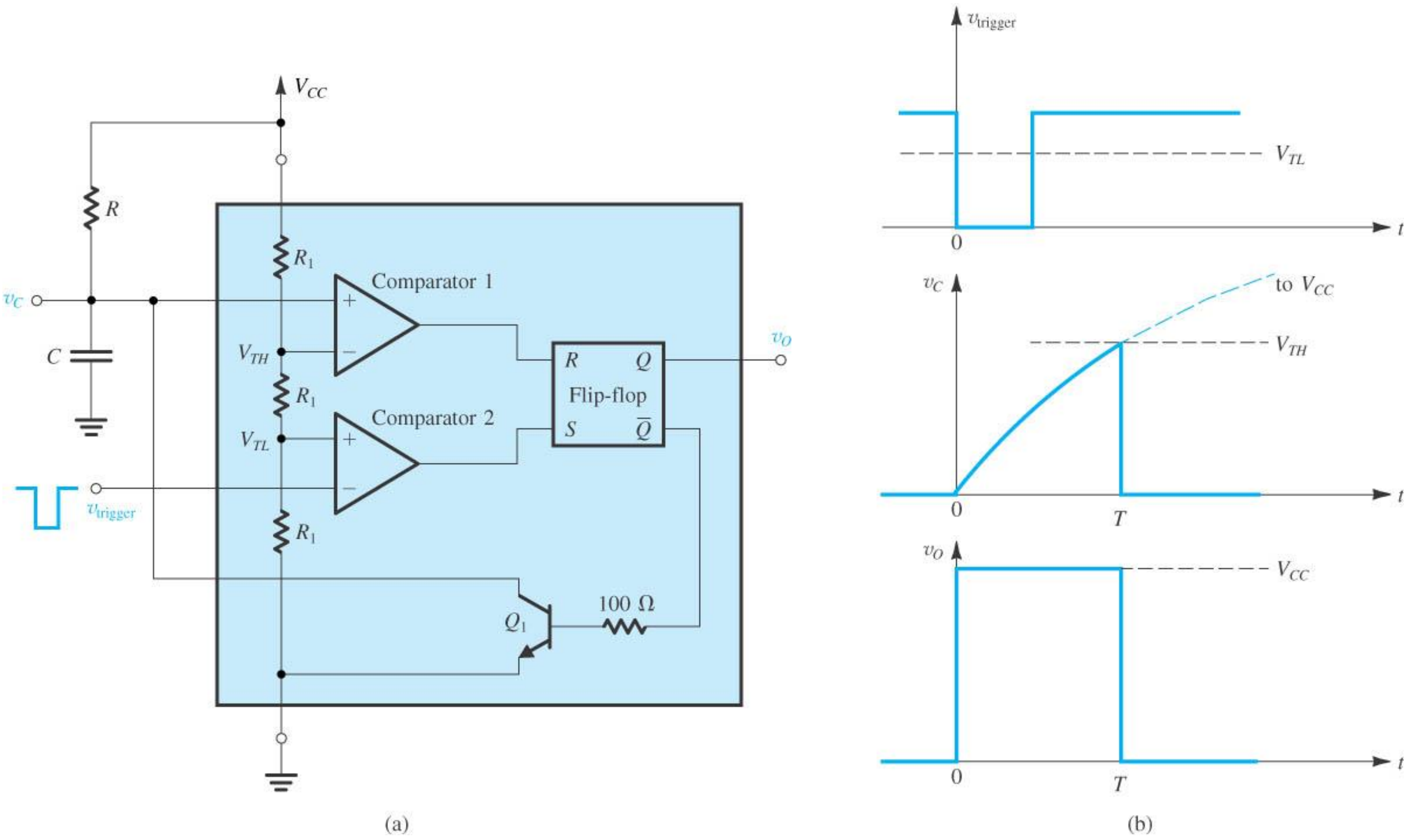
Duty Cycle >50%

Normally the 555 timer is unable to produce a duty cycle of 50% or less. This is due to the fact that the first half of the cycle both Ra and Rb determine the charging interval **(T1)**; where Rb alone determines the discharge interval **(T2)**.

Duty Cycle <50%

To allow a Duty Cycle of 50% or less, a Diode D1 is placed in parallel with Rb such that during the charging cycle (T1) Rb is bypassed. This allows Ra and Rb to act independently, allowing a duty cycle of nearly 0% to nearly 100%.

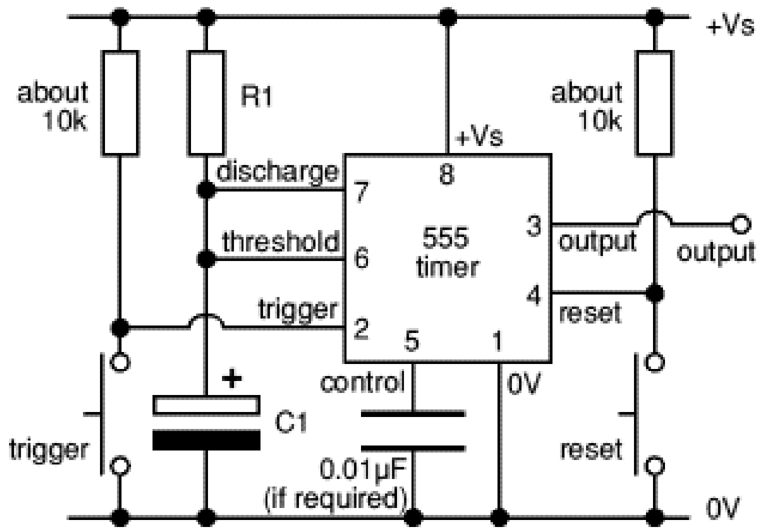
MONO-STABLE OPERATION

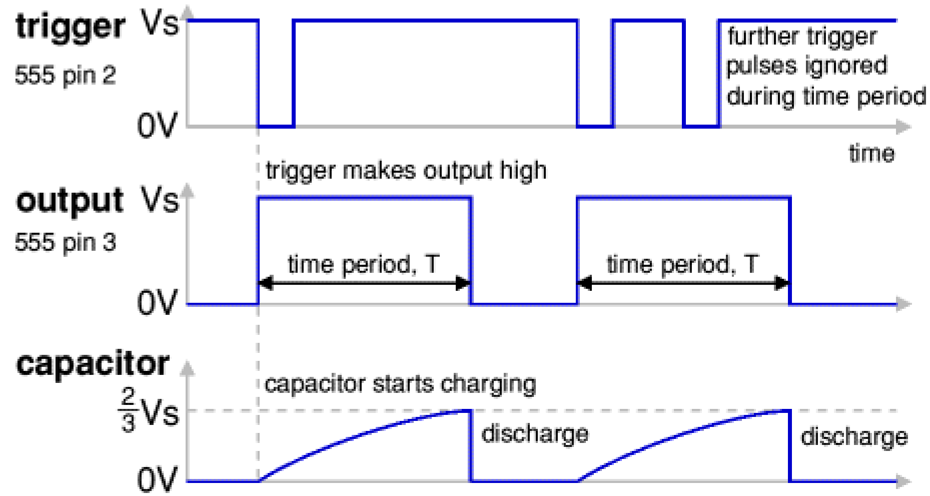


time period, **T = 1.1 × R × C**

MONO-STABLE OPERATION

time period, **T = 1.1 × R1 × C1**

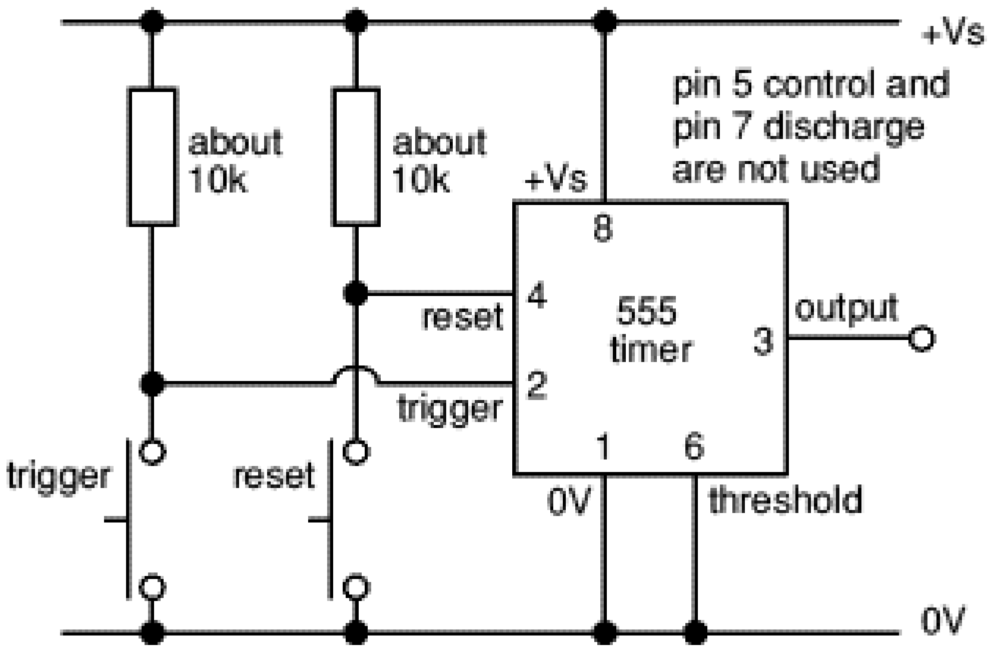




The timing period is triggered (started) when the **trigger** input (555 pin 2) is less than $^1/_3$ Vs, this makes the **output** high (+Vs) and the capacitor C1 starts to charge through resistor R1. Once the time period has started further trigger pulses are ignored. The **threshold** input (555 pin 6) monitors the voltage across C1 and when this reaches $^2/_3$ Vs the time period is over and the **output** becomes low. At the same time **discharge** (555 pin 7) is connected to 0V, discharging the capacitor ready for the next trigger.

The **reset** input (555 pin 4) overrides all other inputs and the timing may be cancelled at any time by connecting reset to 0V, this instantly makes the output low and discharges the capacitor. If the reset function is not required the reset pin should be connected to +Vs.

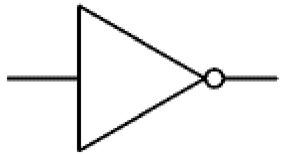
Bistable Operation

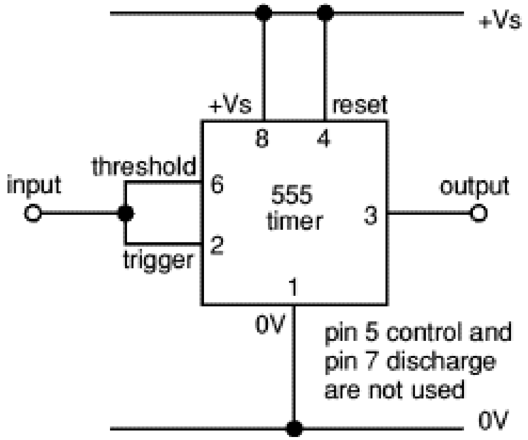


The circuit is called a <u>bi</u>stable because it is stable in **two** states: output high and output low. It is also known as a 'flip-flop'.It has two inputs:

Trigger (555 pin 2) makes the **output high** Trigger is 'active low', it functions when < $^{1}/_{3}$ Vs.
Reset (555 pin 4) makes the **output low** Reset is 'active low', it resets when < 0.7V.

INVERTING BUFFER (SCHMITT TRIGGER) OR NOT GATE





The buffer circuit's input has a very high impedance (about 1M) so it requires only a few µA, but the output can sink or source up to 200mA. This enables a high impedance signal source (such as an LDR) to switch a low impedance output transducer (such as a lamp).It is an **inverting buffer** or **Not Gate** because the output logic state (low/high) is the inverse of the input state:

Input low (< $^1/_3$ Vs) makes **output high**, +Vs

Input high (> $^2/_3$ Vs) makes **output low**, 0V

When the input voltage is between $^1/_3$ and $^2/_3$ Vs the output remains in its present state. This intermediate input region is a deadspace where there is no response, a property called **hysteresis**. This type of circuit is called a **Schmitt trigger**. If high sensitivity is required the hysteresis is a problem, but in many circuits it is a helpful property. It gives the input a high immunity to noise because once the circuit output has switched high or low the input must change back by at least $^1/_3$ Vs to make the output switch back.

Thanks